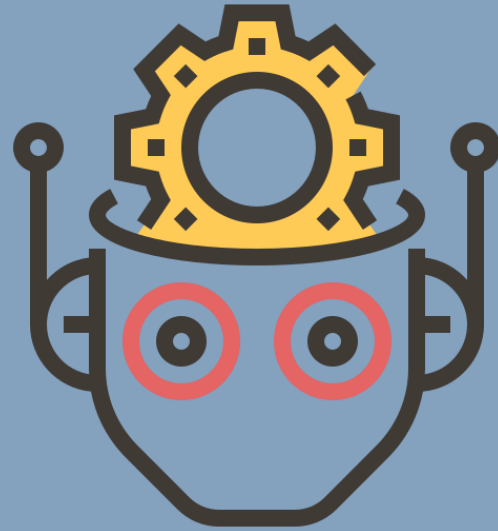


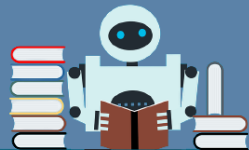


Deep Learning



스마트인재개발원
Smart Human Resources Development

정 봉 군 연구원

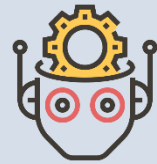


1

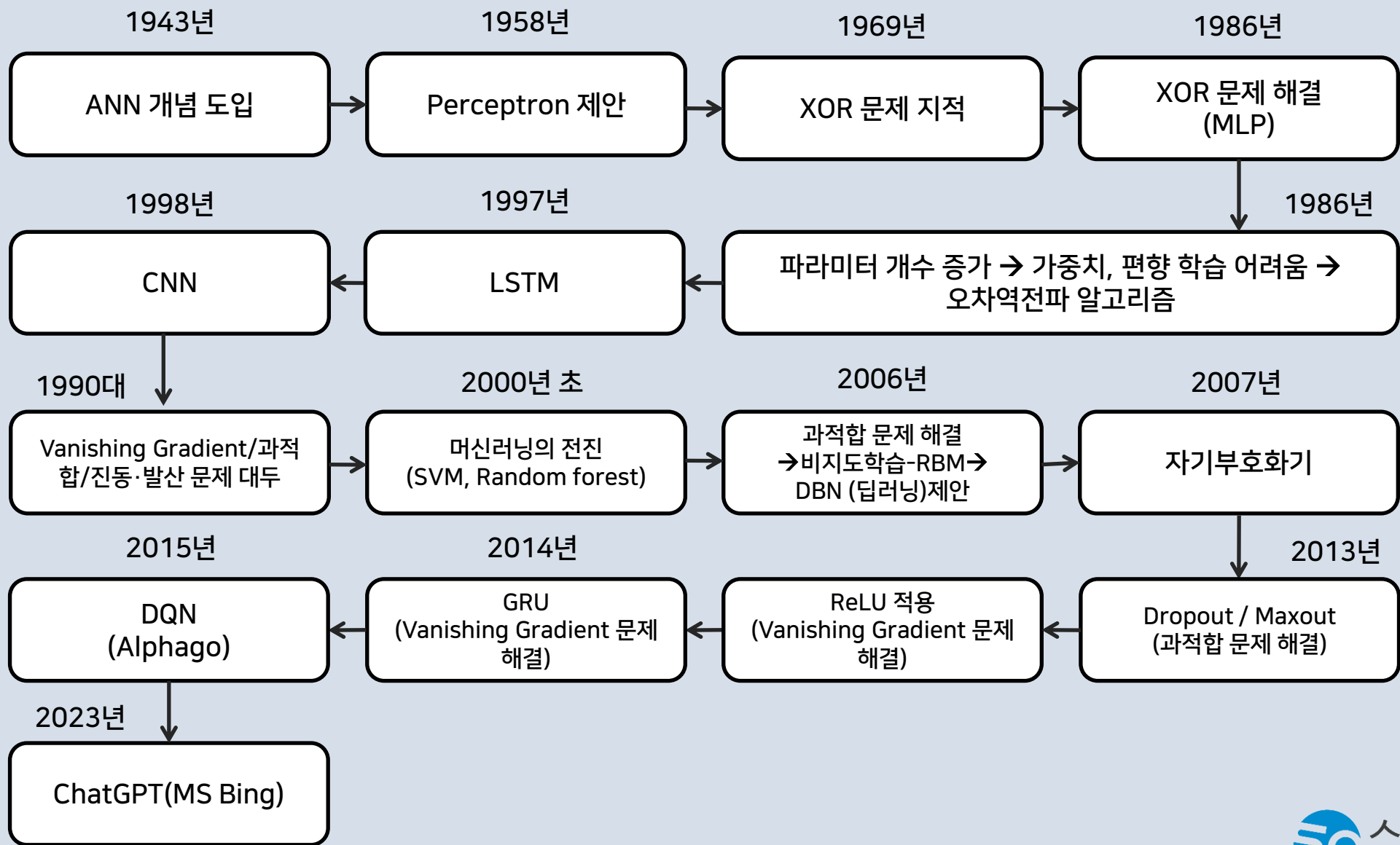
퍼셉트론의 개념을 이해할 수 있다.

2

다층 퍼셉트론의 개념을 이해할 수 있다.



딥러닝 역사



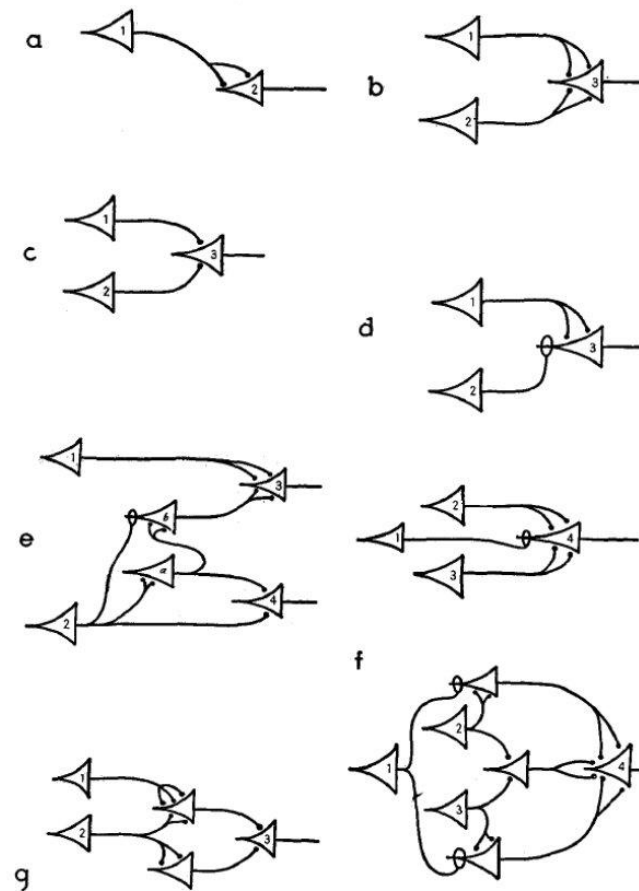
BULLETIN OF
MATHEMATICAL BIOPHYSICS
VOLUME 5, 1943

A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY

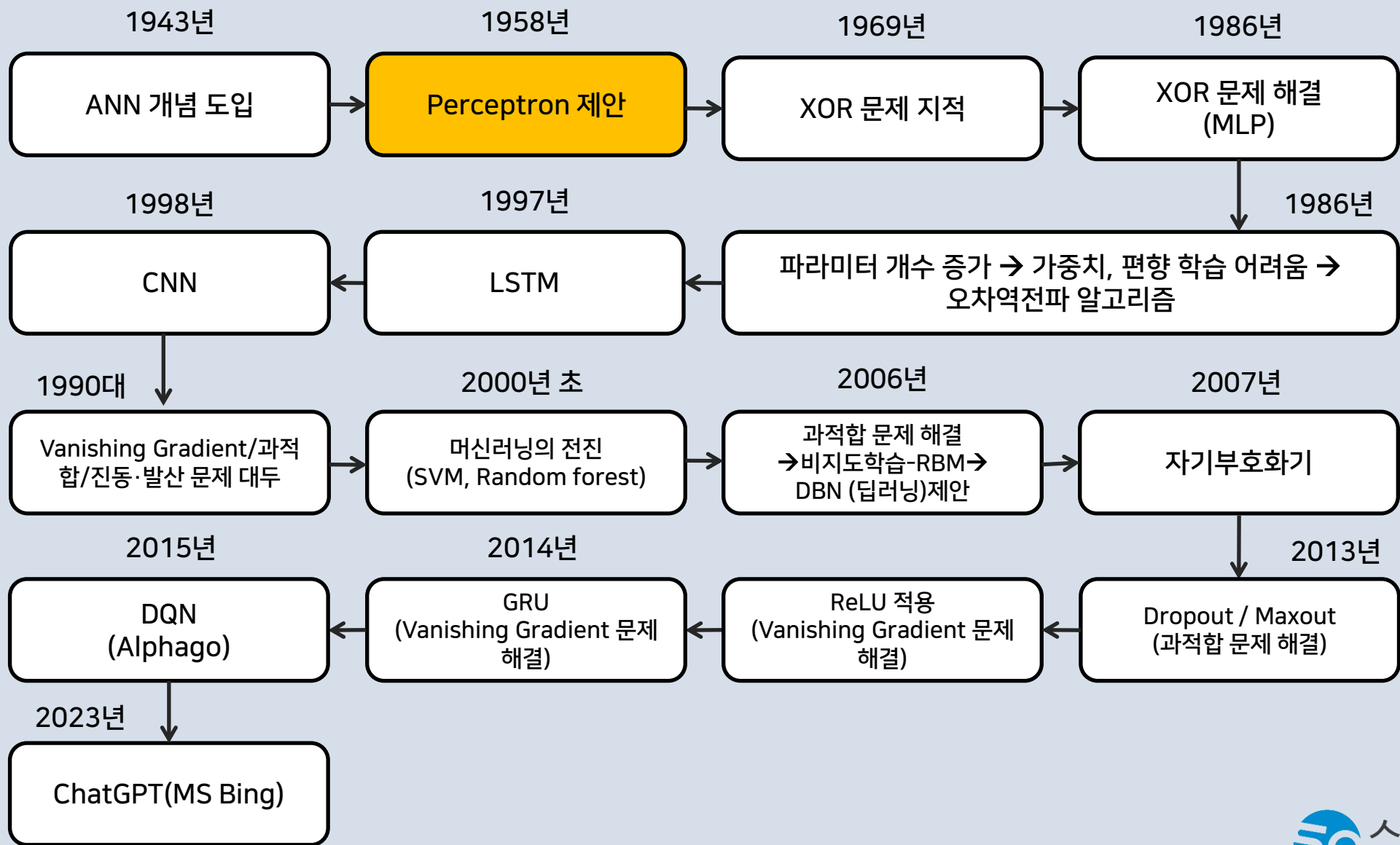
WARREN S. MCCULLOCH AND WALTER PITTS

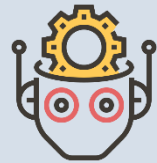
FROM THE UNIVERSITY OF ILLINOIS, COLLEGE OF MEDICINE,
DEPARTMENT OF PSYCHIATRY AT THE ILLINOIS NEUROPSYCHIATRIC INSTITUTE,
AND THE UNIVERSITY OF CHICAGO

Because of the “all-or-none” character of nervous activity, neural events and the relations among them can be treated by means of propositional logic. It is found that the behavior of every net can be described in these terms, with the addition of more complicated logical means for nets containing circles; and that for any logical expression satisfying certain conditions, one can find a net behaving in the fashion it describes. It is shown that many particular choices among possible neurophysiological assumptions are equivalent, in the sense that for every net behaving under one assumption, there exists another net which behaves under the other and gives the same results, although perhaps not in the same time. Various applications of the calculus are discussed.



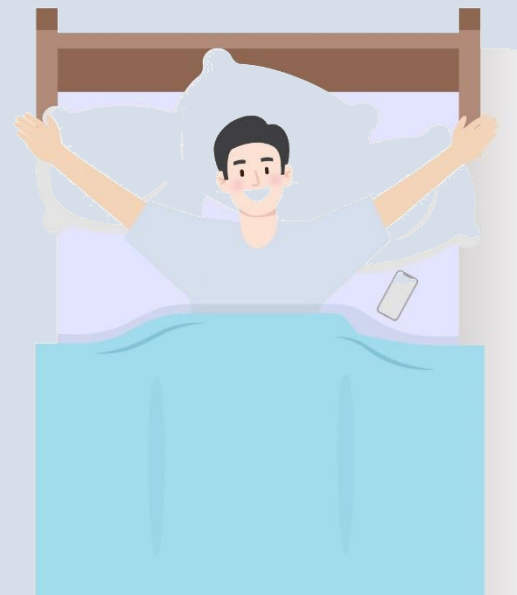
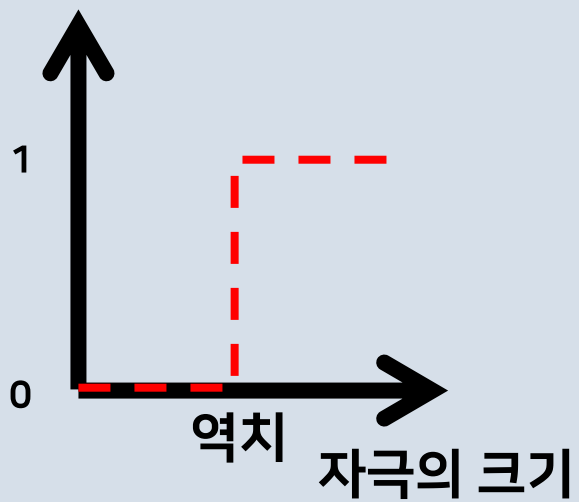
출처 : 논문_신경 활동에 있어서 관념의 논리적 미적분

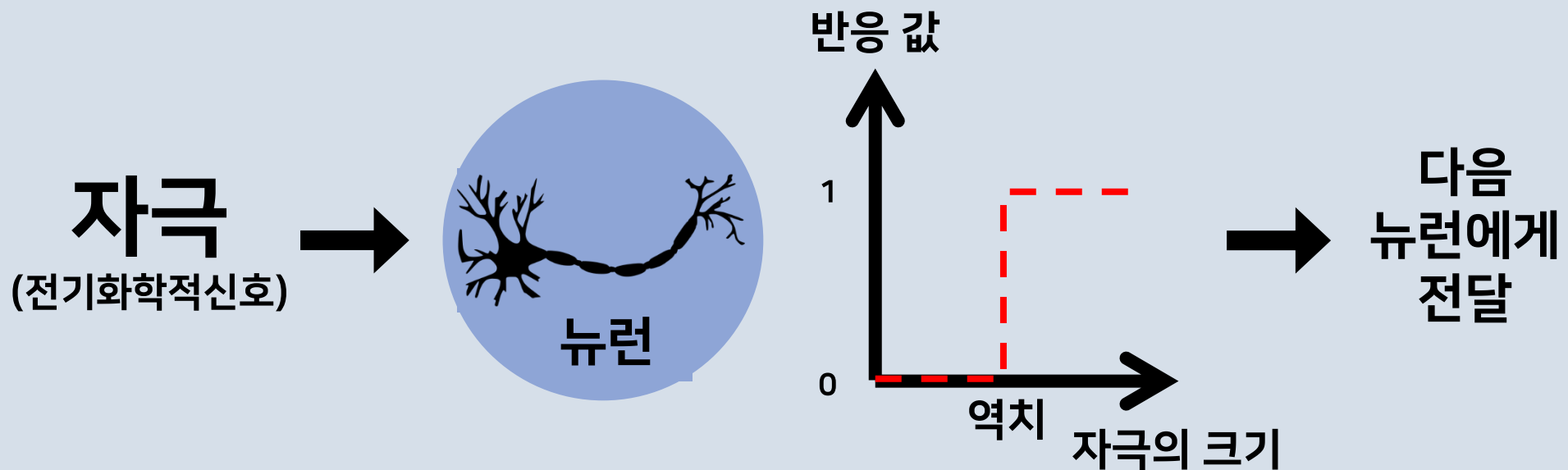




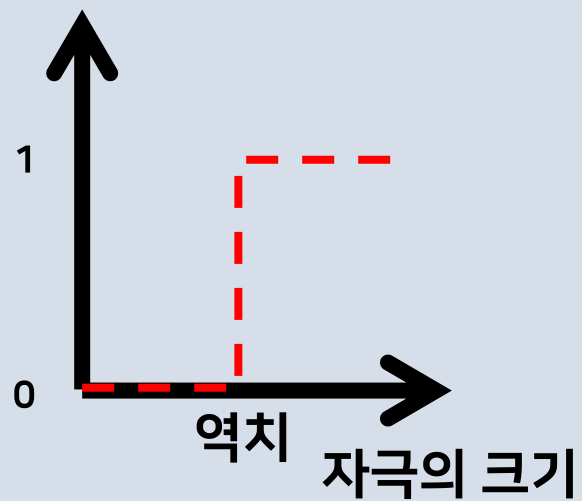
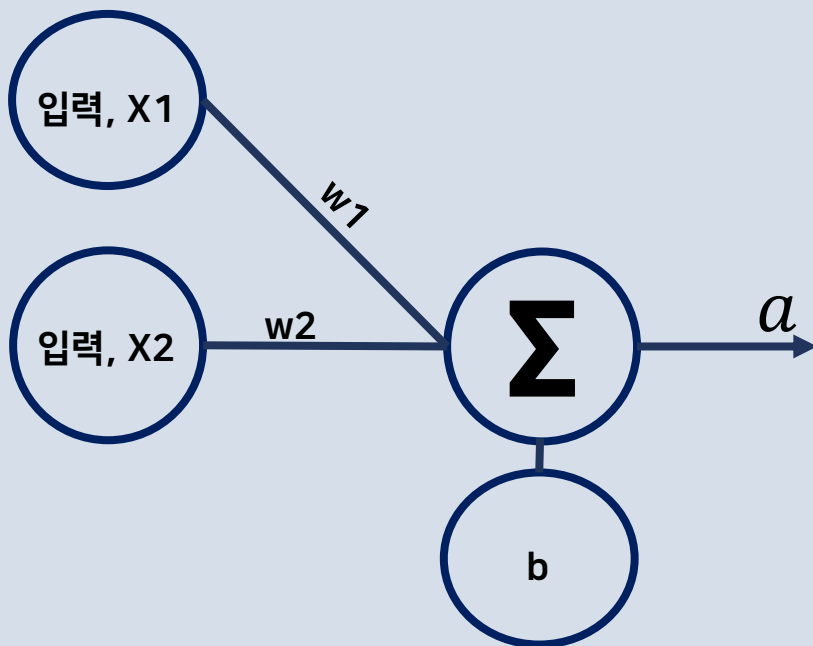
퍼셉트론(Perceptron)

퍼셉트론(Perceptron)

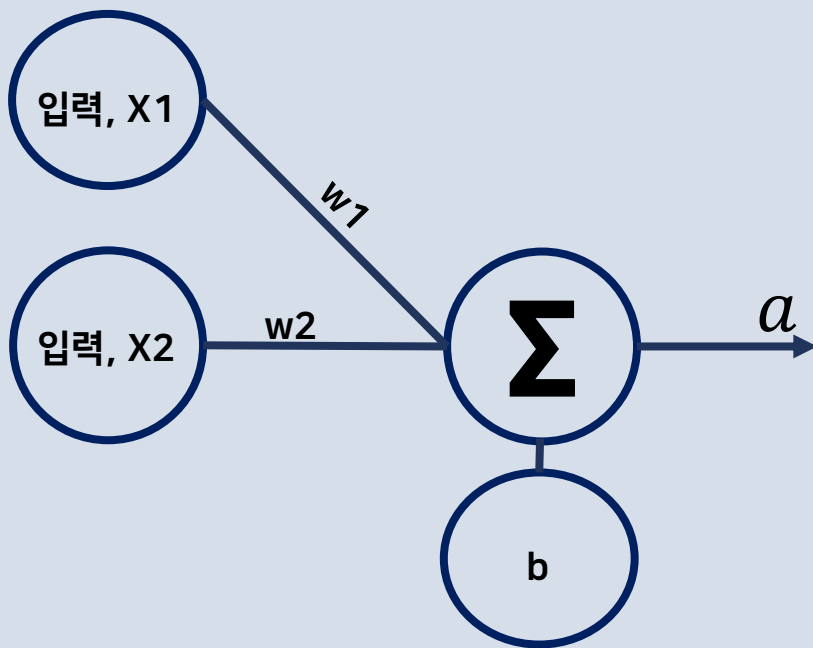




어떠한 자극에 대해서 **일정 자극 이상(역치 이상)**이 되면,
다음 뉴런에게 자극을 전달

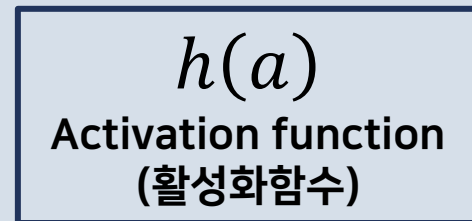


출력 y

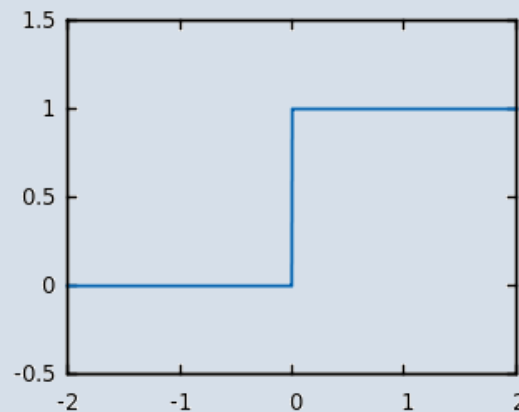


$$a = W_1X_1 + W_2X_2 + b$$

$$y = h(a)$$



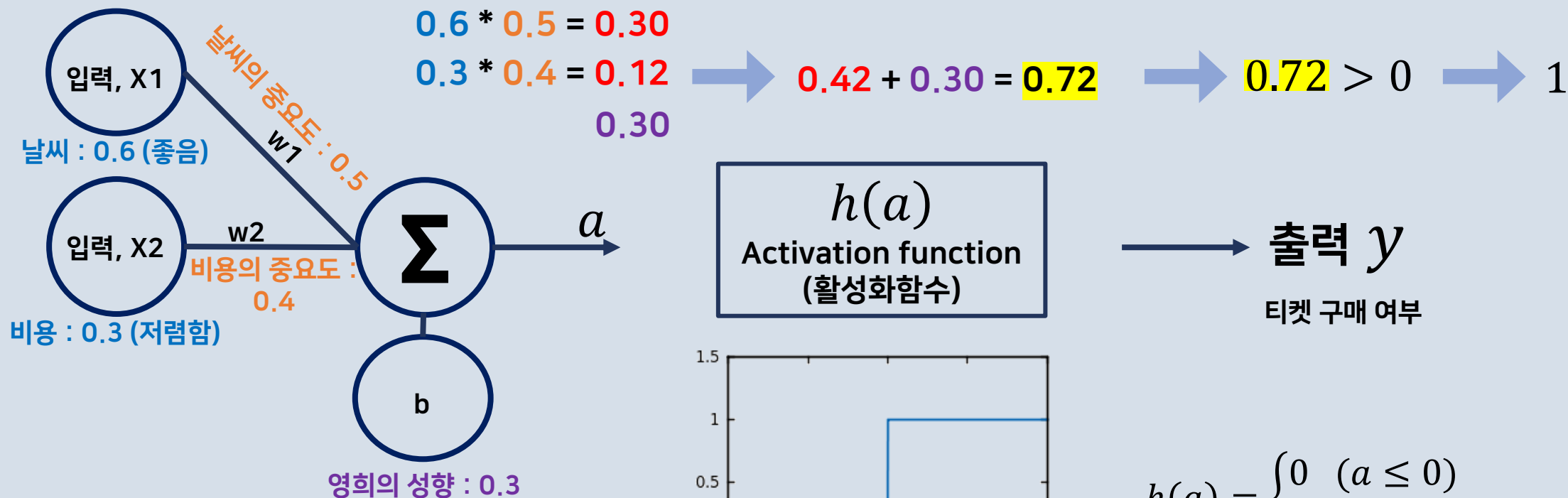
출력 y



$$h(a) = \begin{cases} 0 & (a \leq 0) \\ 1 & (a > 0) \end{cases}$$

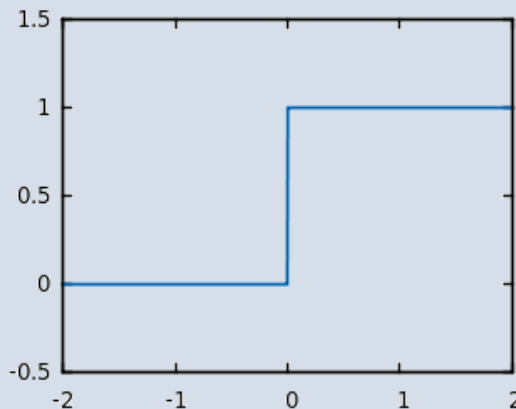
Step function(계단함수)

ex) 콘서트장 예매 - 영희의 경우



$$a = W_1X_1 + W_2X_2 + b$$

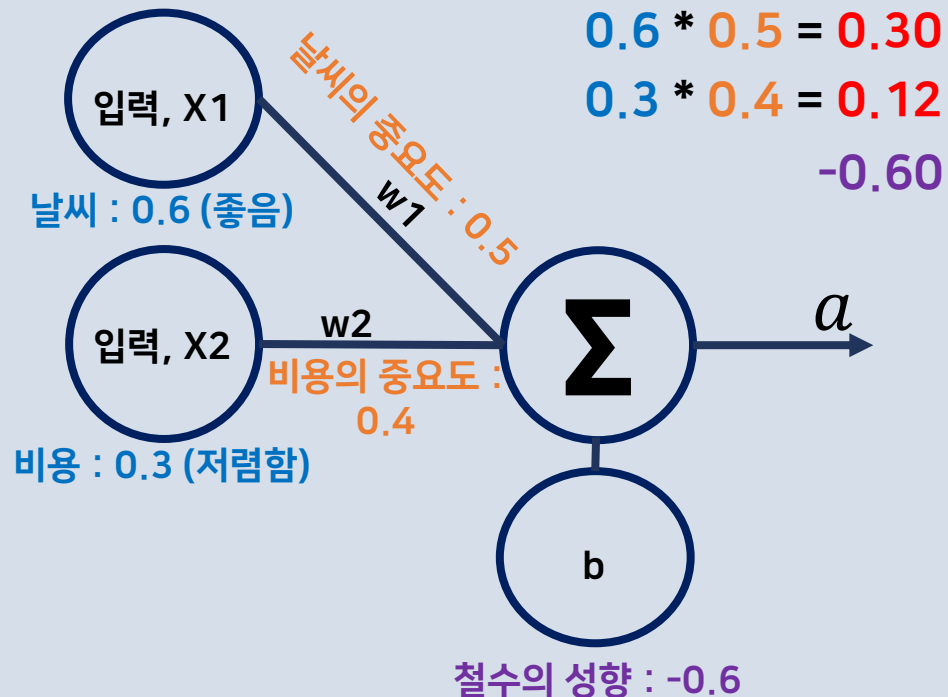
$$y = h(a)$$



$$h(a) = \begin{cases} 0 & (a \leq 0) \\ 1 & (a > 0) \end{cases}$$

Step function(계단함수)

ex) 콘서트장 예매 - 철수의 경우

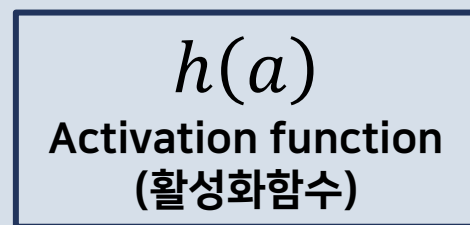


$$0.6 * 0.5 = 0.30$$

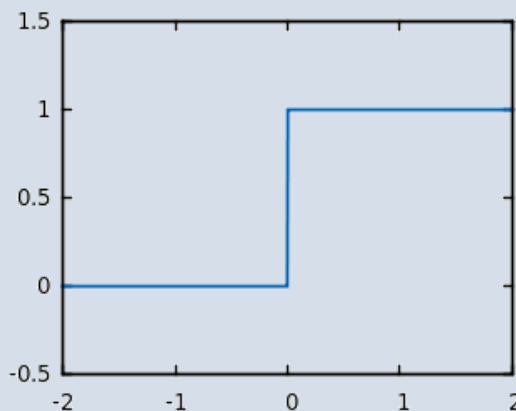
$$0.3 * 0.4 = 0.12$$

$$-0.60$$

$$0.42 - 0.60 = -0.18 \rightarrow -0.18 \leq 0 \rightarrow 0$$



출력 y
티켓 구매 여부



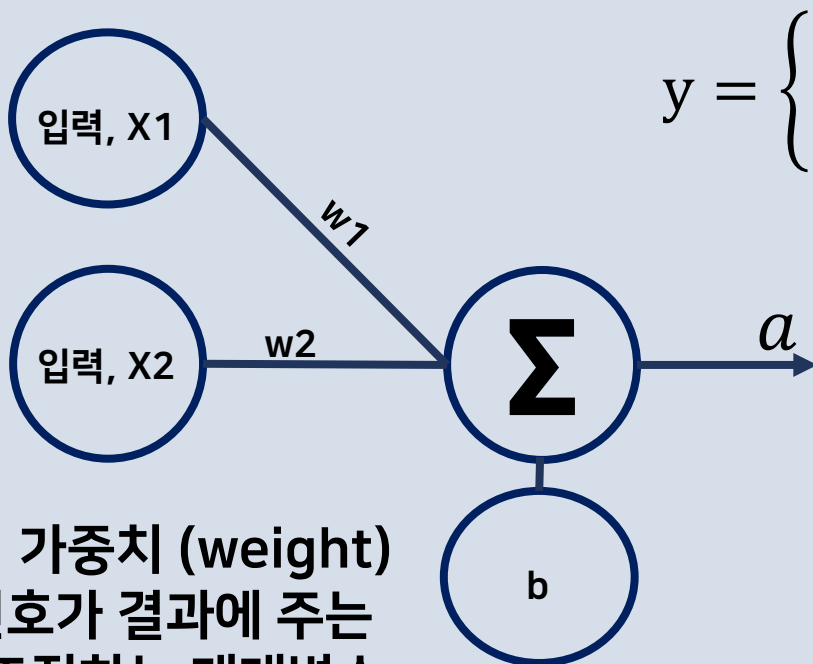
$$h(a) = \begin{cases} 0 & (a \leq 0) \\ 1 & (a > 0) \end{cases}$$

$$a = W_1X_1 + W_2X_2 + b$$

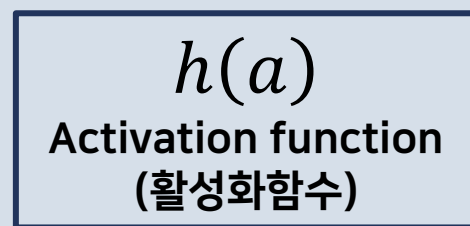
$$y = h(a)$$

Step function(계단함수)

퍼셉트론(단층)의 작동방식



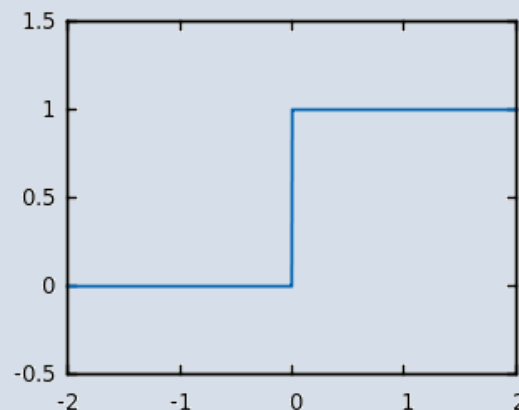
$$y = \begin{cases} 0, & (W_1X_1 + W_2X_2 + b \leq 0) \\ 1, & (W_1X_1 + W_2X_2 + b > 0) \end{cases}$$



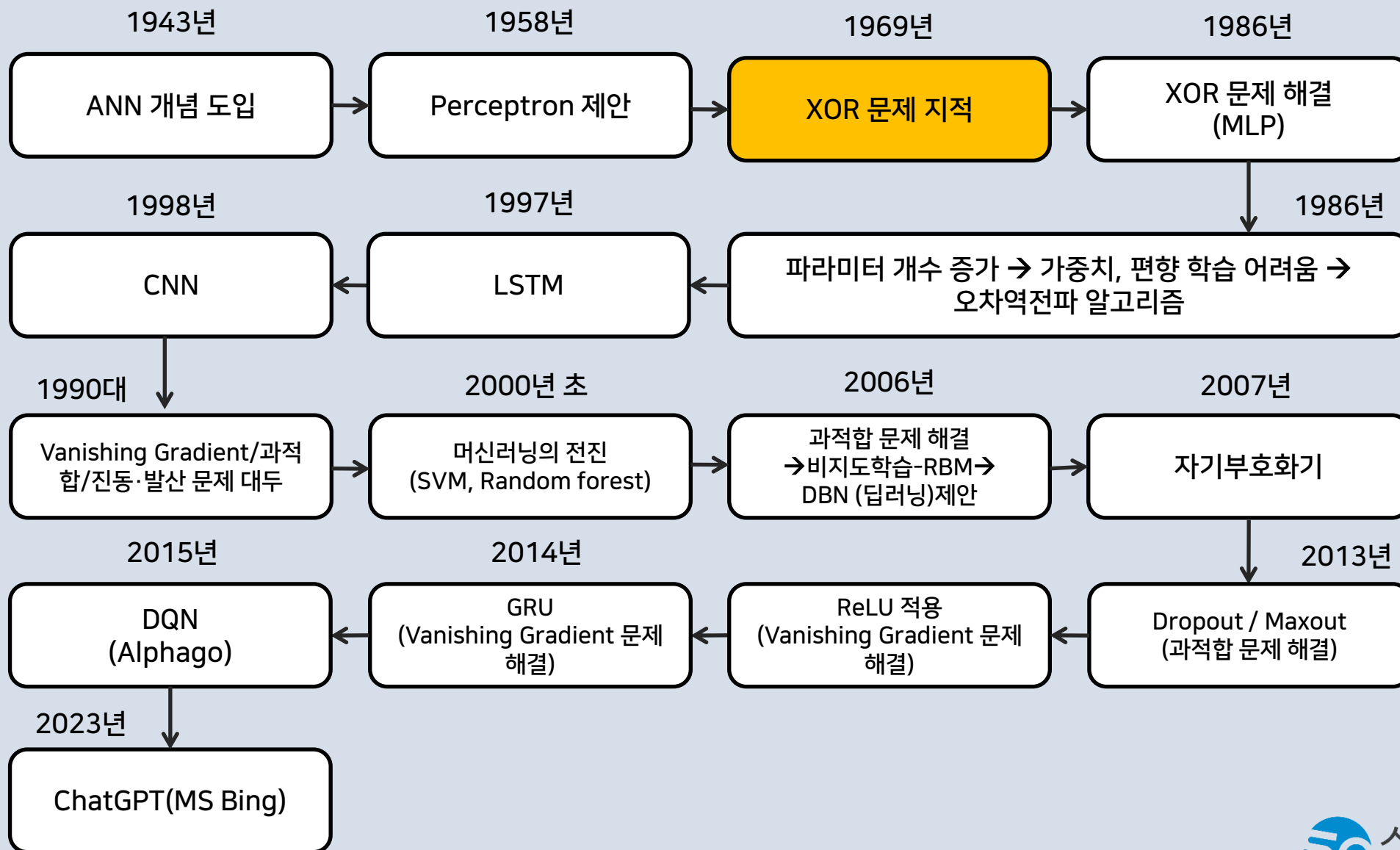
출력 y

W_1, W_2 : 가중치 (weight)
각 입력 신호가 결과에 주는
영향력을 조절하는 매개변수

b : 편향 (bias)
뉴런이 얼마나 쉽게
활성화하느냐를 조절하는 매개변수



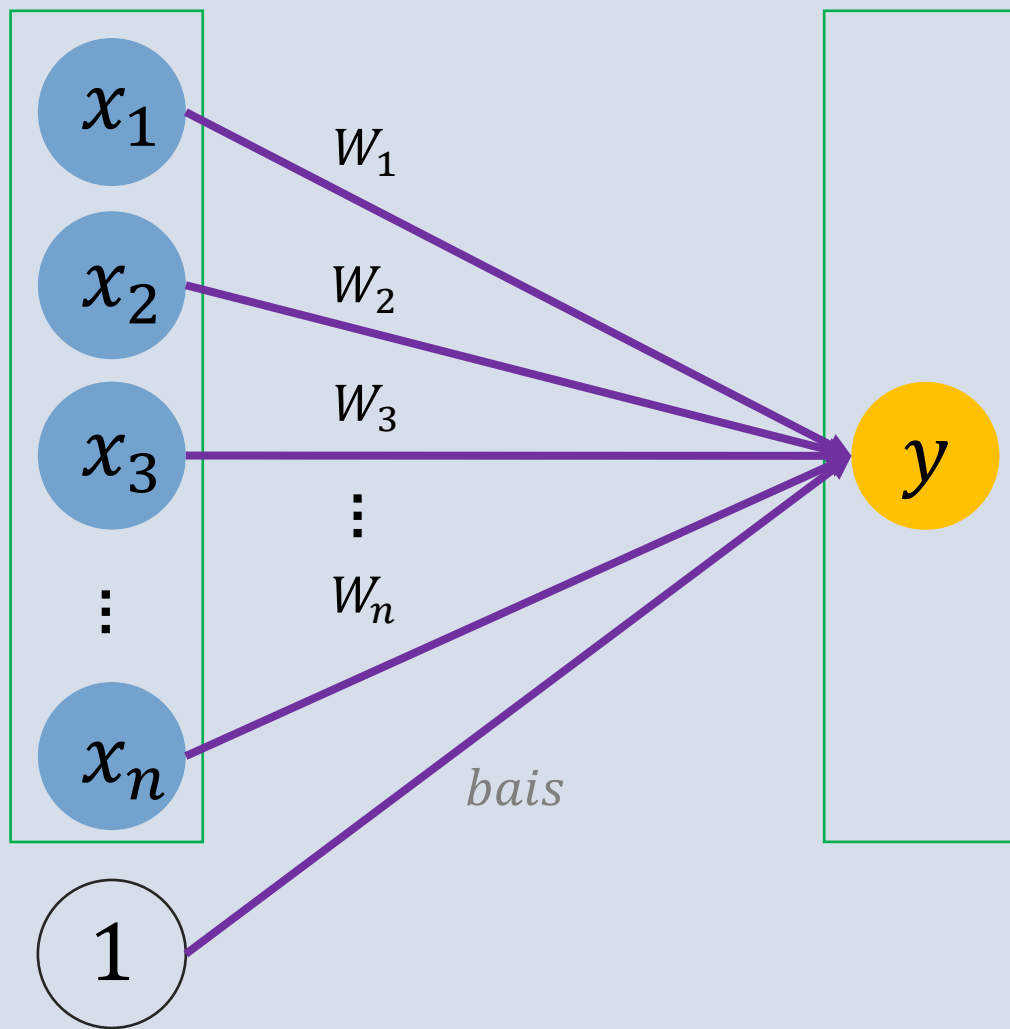
Step function(계단함수)

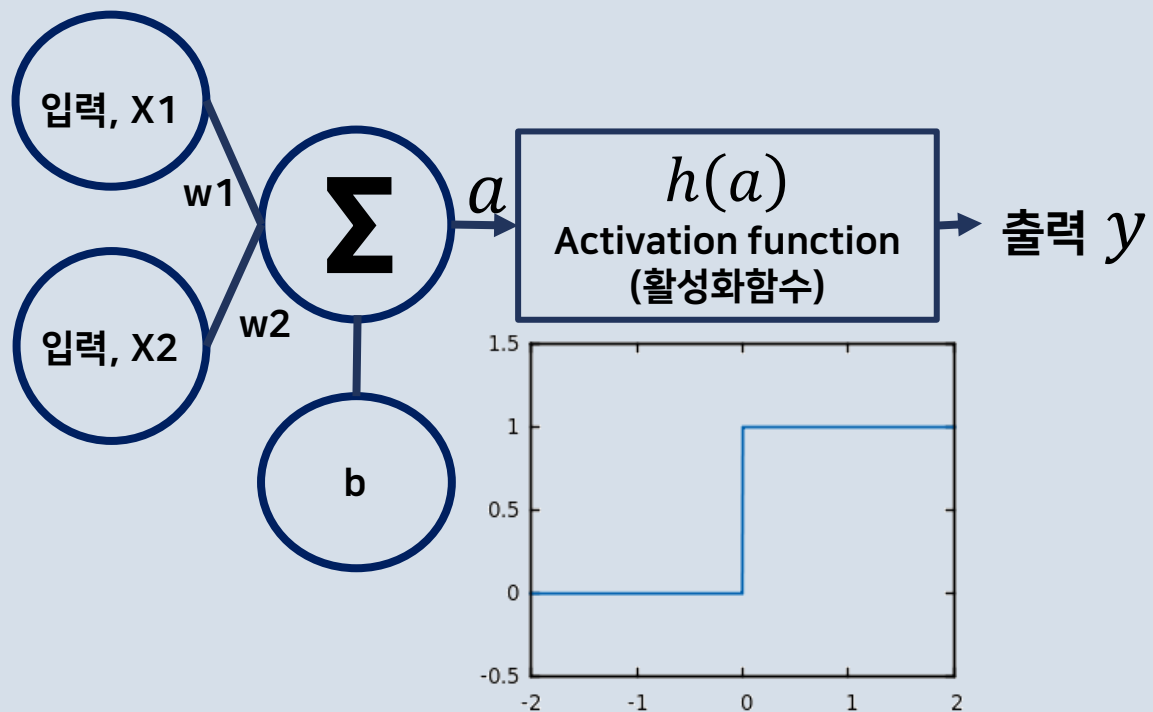


단층 퍼셉트론의 한계 - XOR 문제

입력층(input layer)

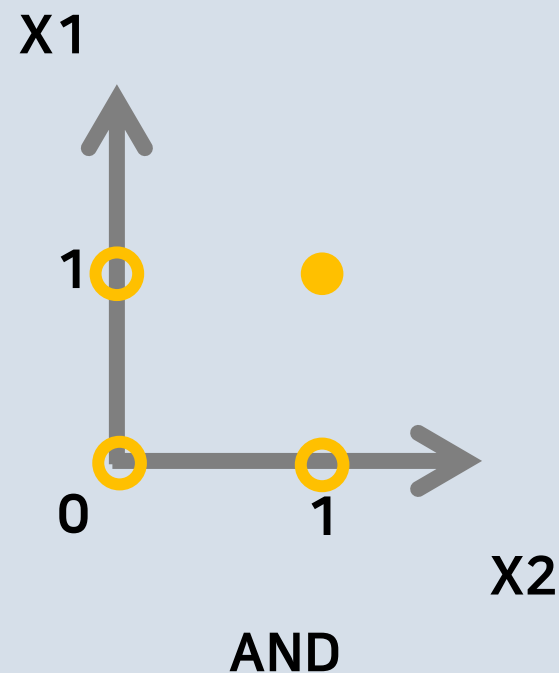
출력층(output layer)



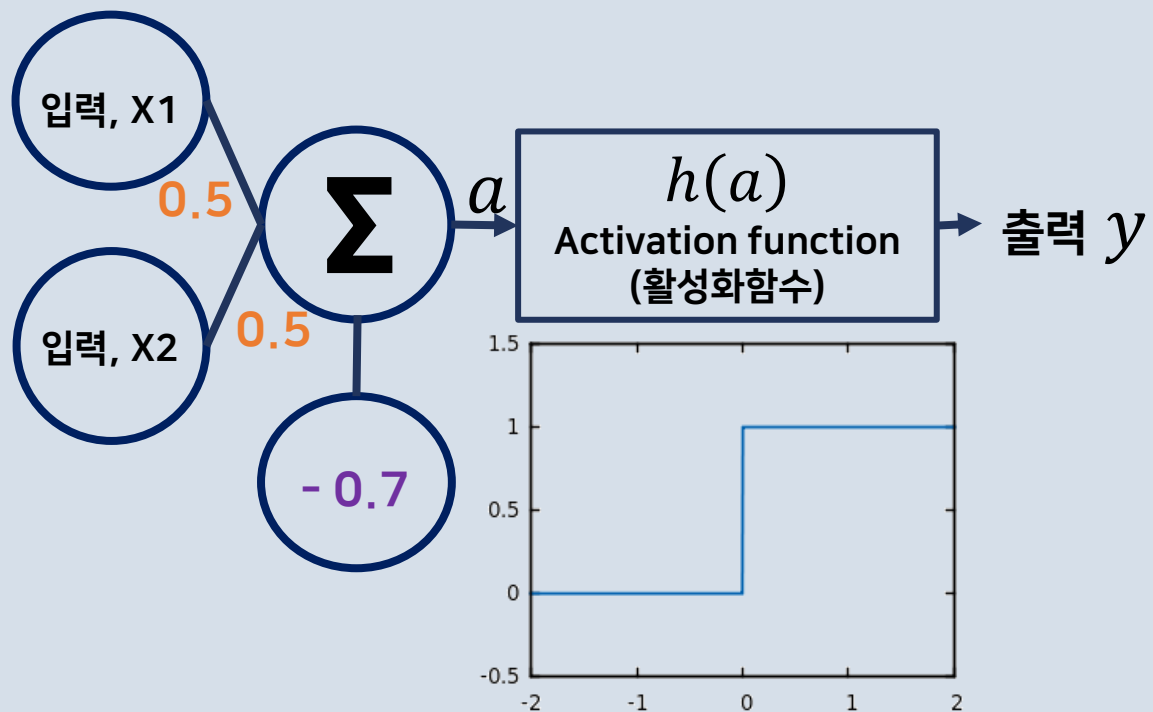


AND 게이트

X1	X2	AND
0	0	0
0	1	0
1	0	0
1	1	1

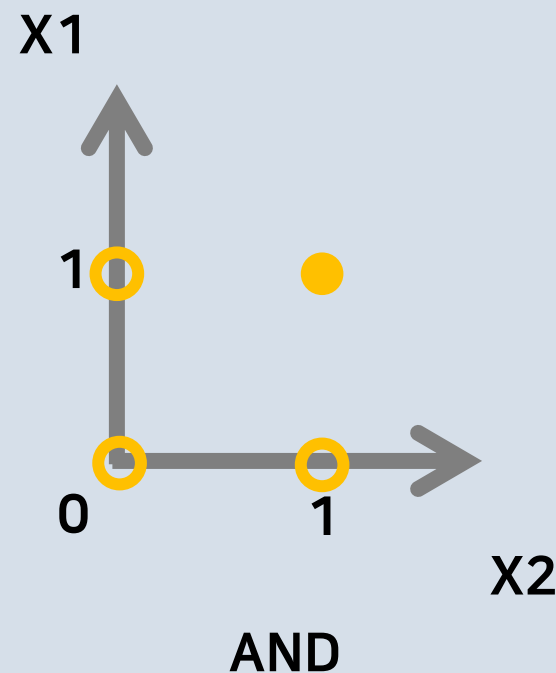


$$y = \begin{cases} 0, & (W_1X_1 + W_2X_2 + b \leq 0) \\ 1, & (W_1X_1 + W_2X_2 + b > 0) \end{cases}$$



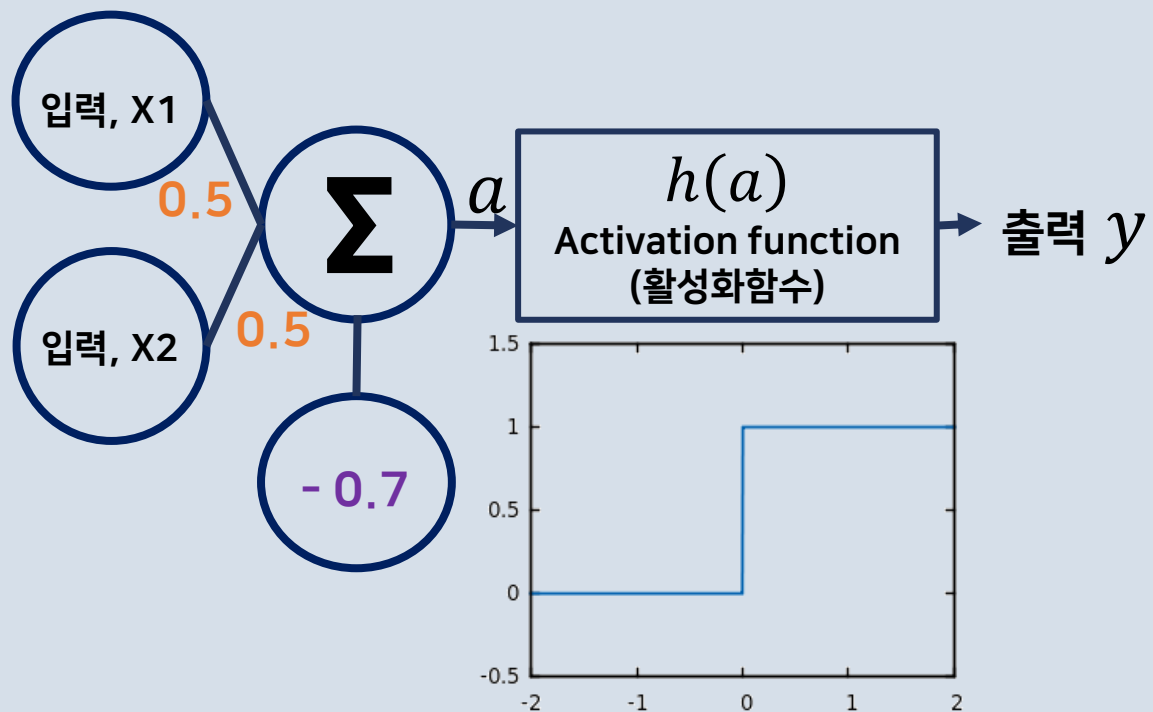
AND 게이트

X1	X2	AND
0	0	0
0	1	0
1	0	0
1	1	1



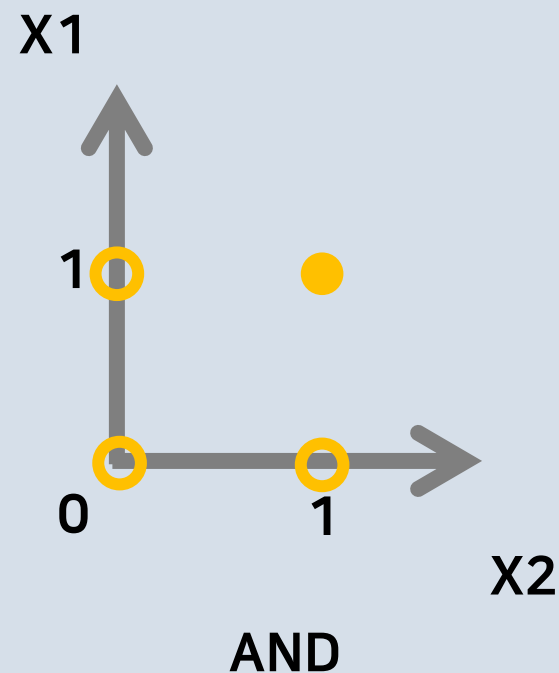
$$0.0 * 0.5 = 0.00$$

$$0.0 * 0.5 = 0.00 \quad -0.70 \quad \longrightarrow \quad 0.00 - 0.70 = -0.70 \quad \longrightarrow \quad -0.70 \leq 0 \quad \longrightarrow \quad 0$$



AND 게이트

X1	X2	AND
0	0	0
0	1	0
1	0	0
1	1	1

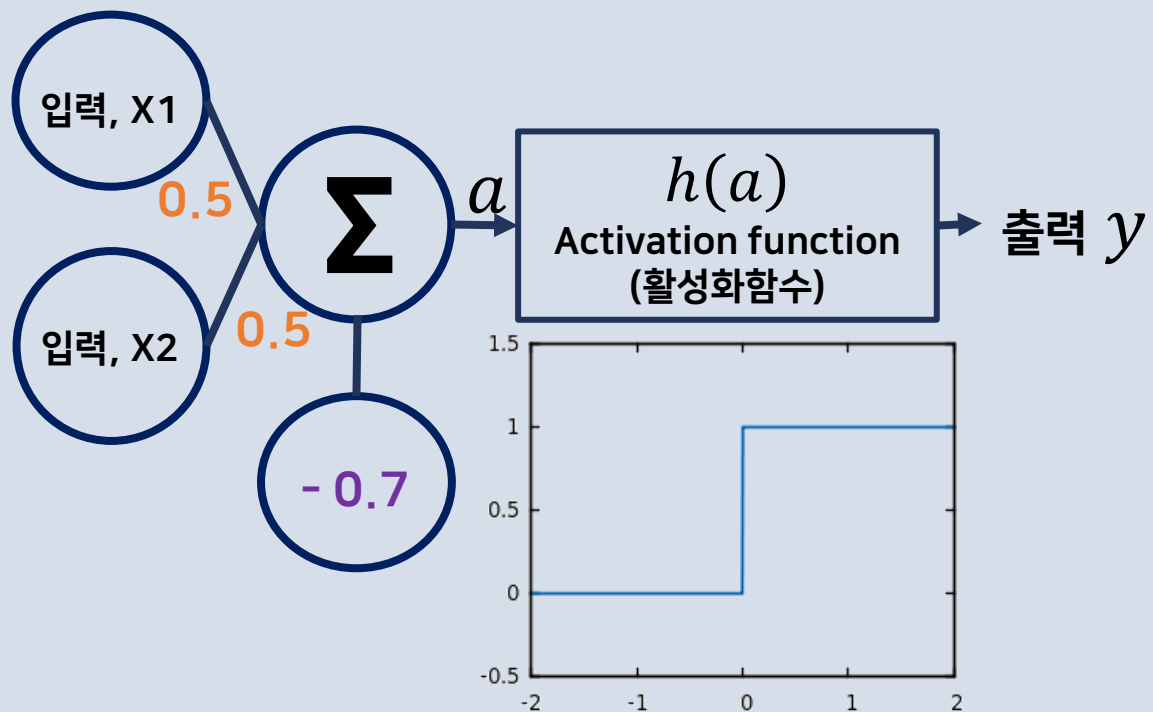


$$0.0 * 0.5 = 0.00$$

$$1.0 * 0.5 = 0.50$$

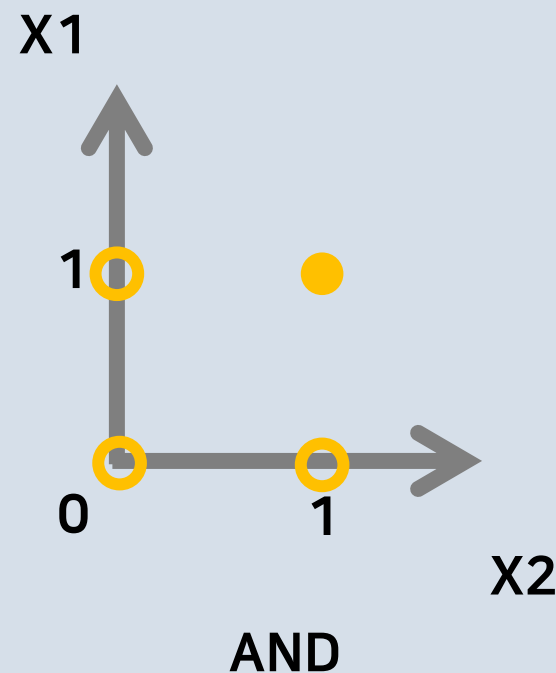
$$-0.70$$

$$\rightarrow 0.50 - 0.70 = -0.20 \rightarrow -0.20 \leq 0 \rightarrow 0$$



AND 게이트

X1	X2	AND
0	0	0
0	1	0
1	0	0
1	1	1

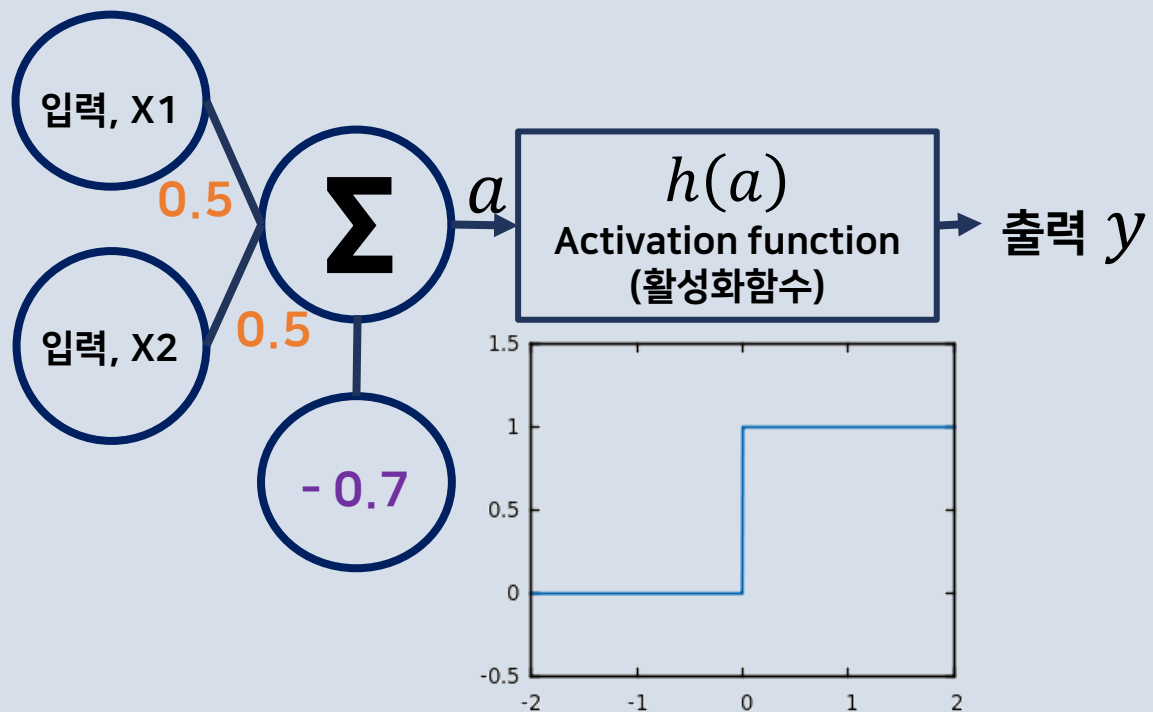


$$1.0 * 0.5 = 0.50$$

$$0.0 * 0.5 = 0.00$$

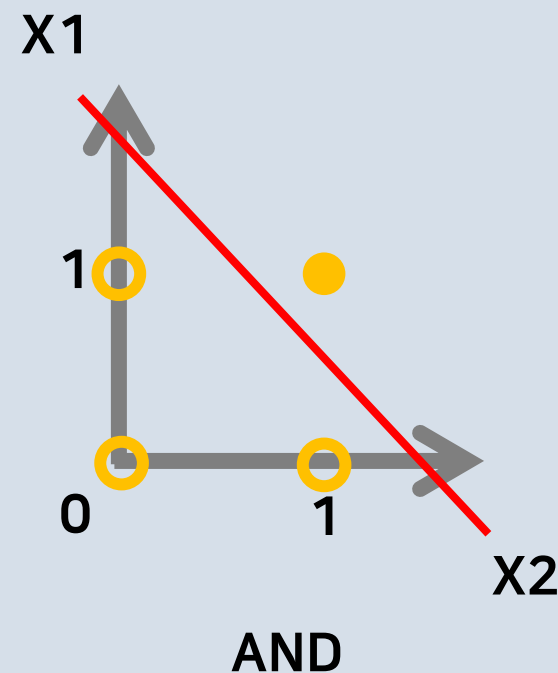
$$-0.70$$

$$\rightarrow 0.50 - 0.70 = -0.20 \rightarrow -0.20 \leq 0 \rightarrow 0$$



AND 게이트

X1	X2	AND
0	0	0
0	1	0
1	0	0
1	1	1



$$1.0 * 0.5 = 0.50$$

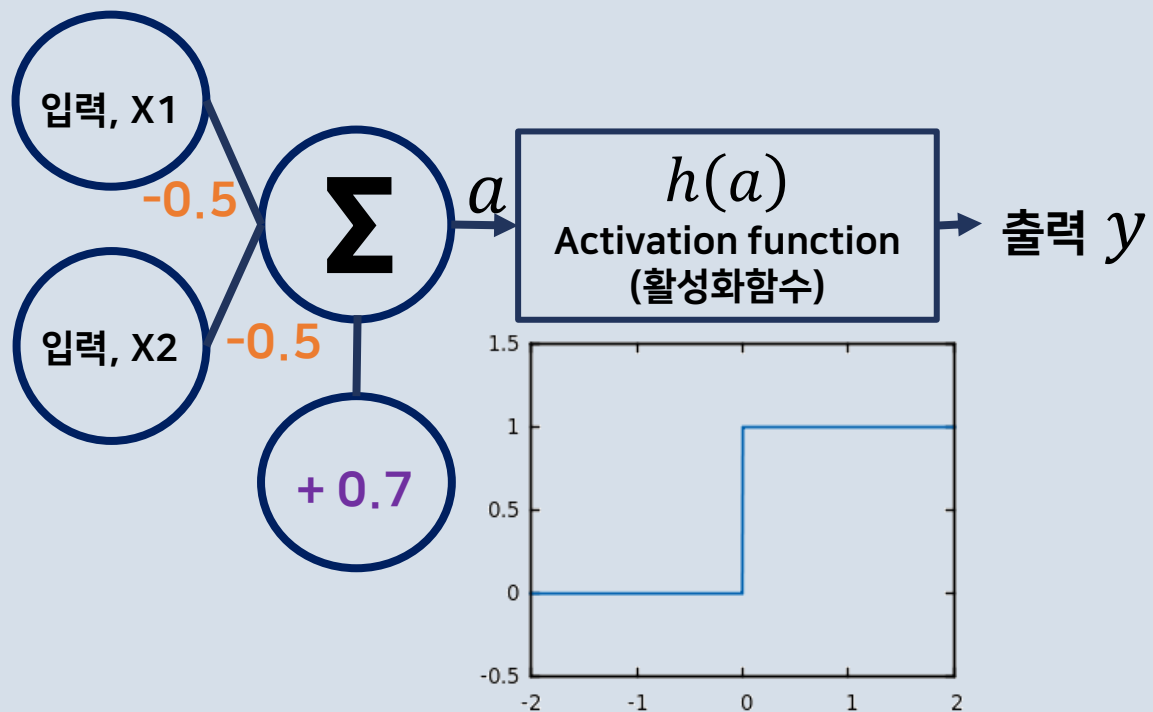
$$1.0 * 0.5 = 0.50$$

$$-0.70$$

$$1.00 - 0.70 = 0.30$$

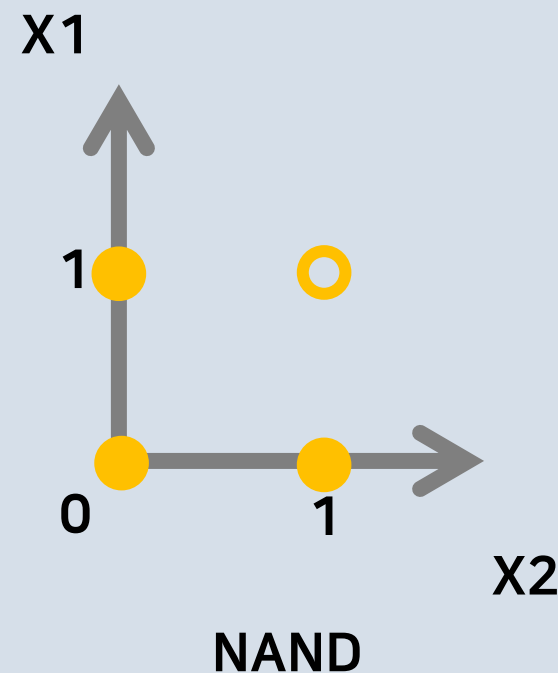
$$0.30 > 0$$

$$\rightarrow 1$$



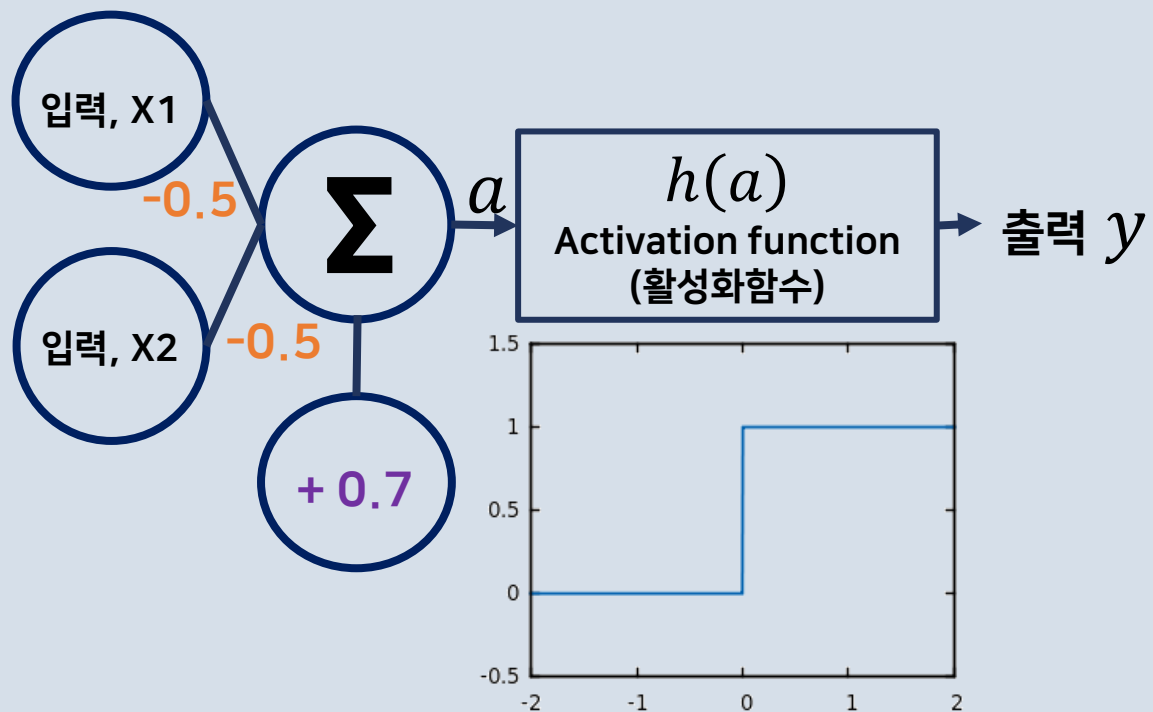
NAND 게이트

X1	X2	NAND
0	0	1
0	1	1
1	0	1
1	1	0



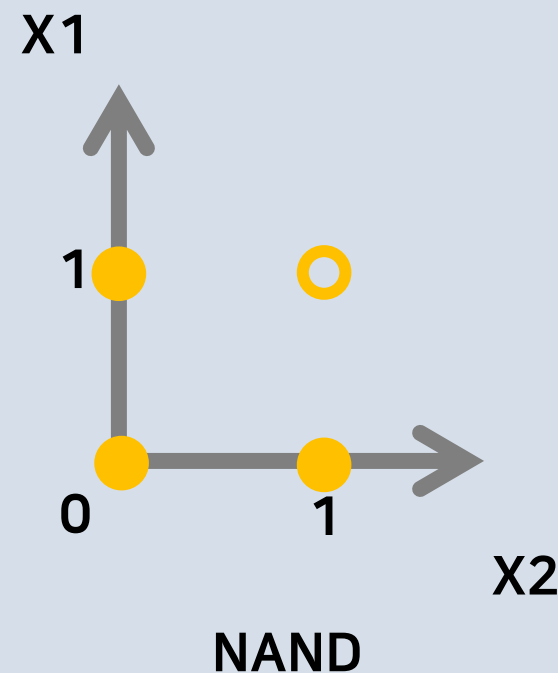
$$0.0 * -0.5 = 0.00$$

$$0.0 * -0.5 = 0.00 \quad + 0.70 = 0.70 \quad \rightarrow \quad 0.70 > 0 \quad \rightarrow \quad 1$$

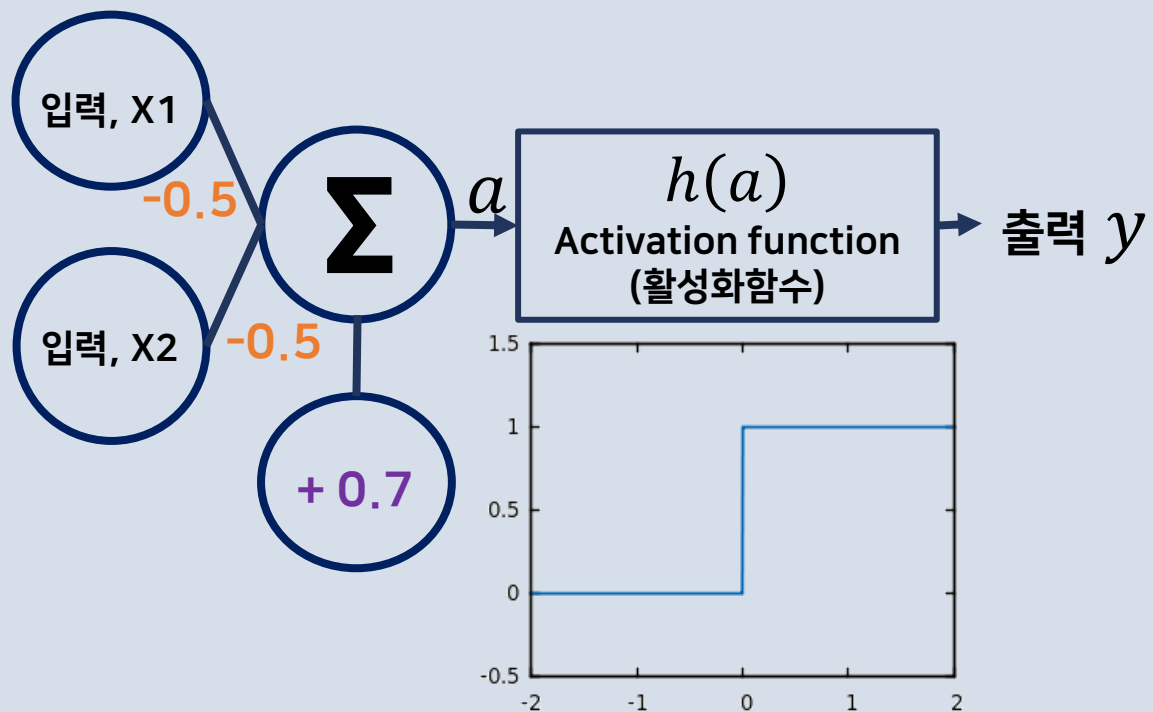


NAND 게이트

X1	X2	NAND
0	0	1
0	1	1
1	0	1
1	1	0

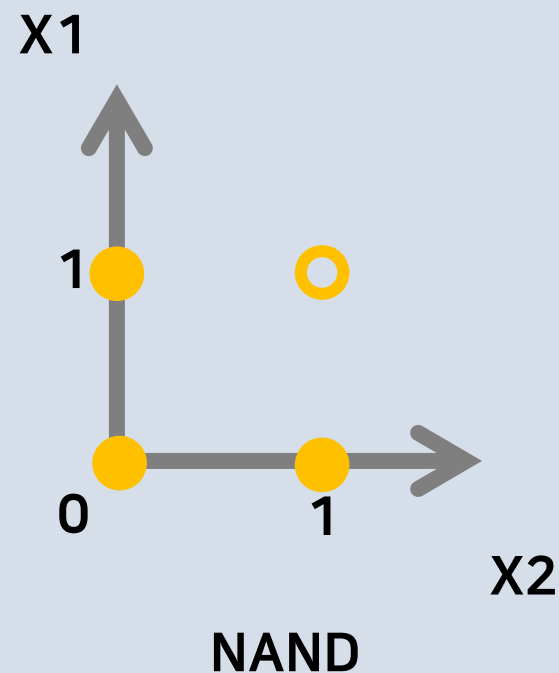


$$\begin{aligned}
 &0.0 * -0.5 = 0.00 \\
 &1.0 * -0.5 = -0.50 \\
 &\quad + 0.70 \\
 &\rightarrow -0.50 + 0.70 = 0.20 \rightarrow 0.20 \geq 0 \rightarrow 1
 \end{aligned}$$



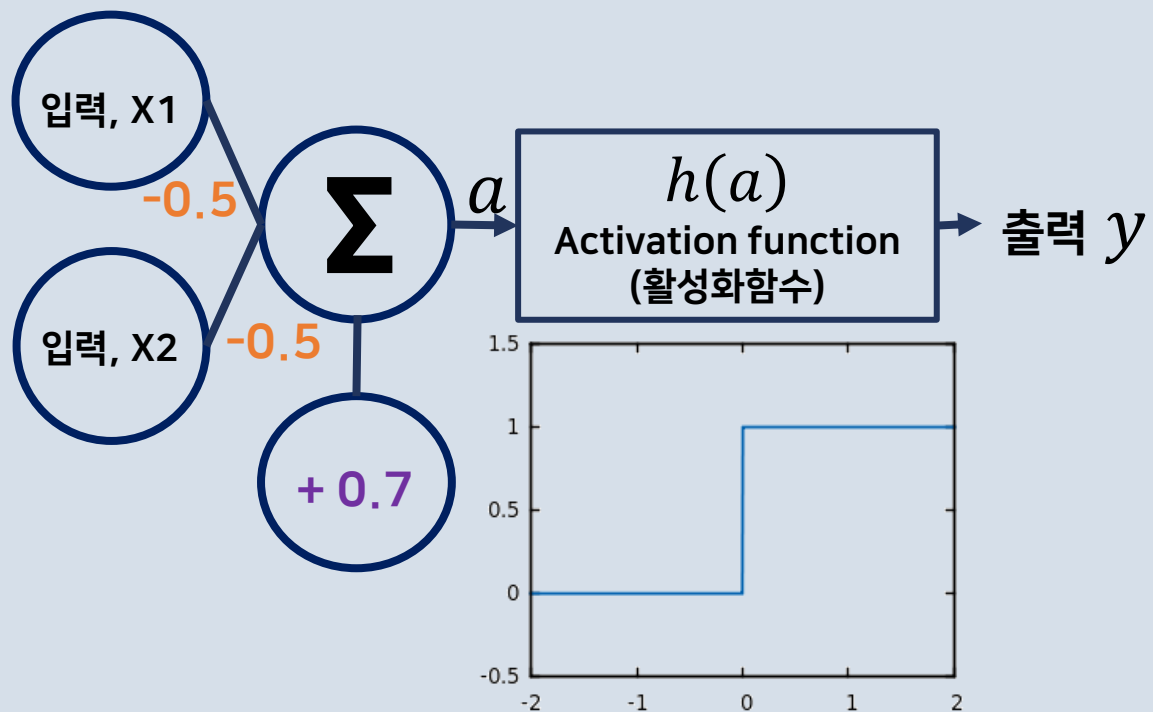
NAND 게이트

X1	X2	NAND
0	0	1
0	1	1
1	0	1
1	1	0



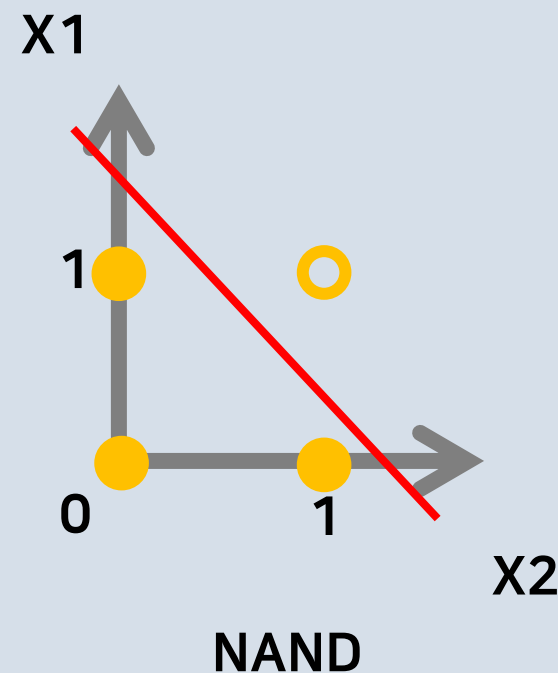
$$1.0 * -0.5 = -0.50$$

$$0.0 * -0.5 = 0.00 \quad \rightarrow \quad -0.50 + 0.70 = 0.20 \quad \rightarrow \quad 0.20 > 0 \quad \rightarrow \quad 1$$



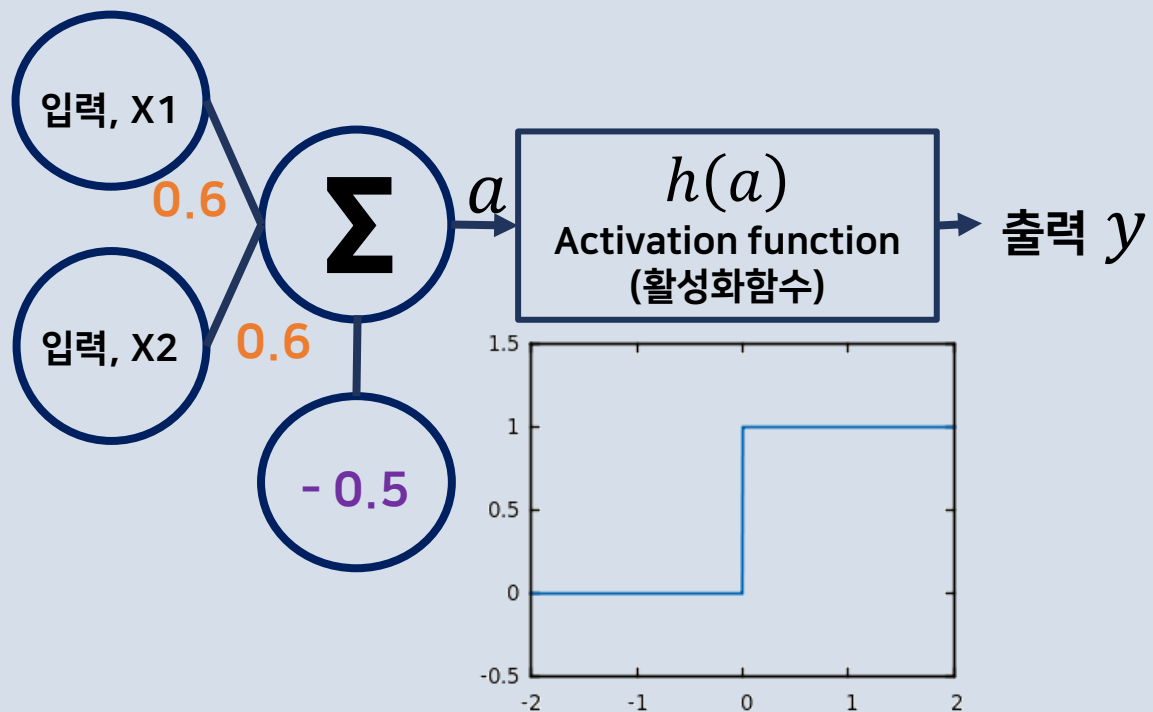
NAND 게이트

X1	X2	NAND
0	0	1
0	1	1
1	0	1
1	1	0



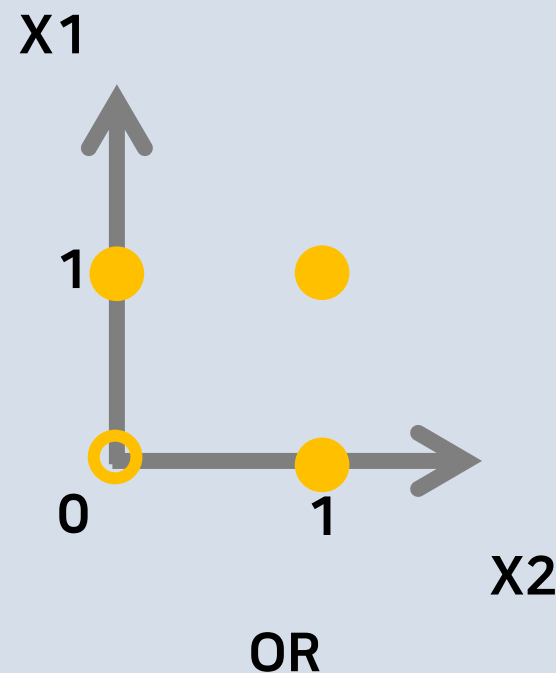
$$1.0 * -0.5 = -0.50$$

$$1.0 * -0.5 = -0.50 \xrightarrow{+0.70} -1.00 + 0.70 = -0.30 \xrightarrow{-0.30 \leq 0} 0$$



OR 게이트

X1	X2	OR
0	0	0
0	1	1
1	0	1
1	1	1



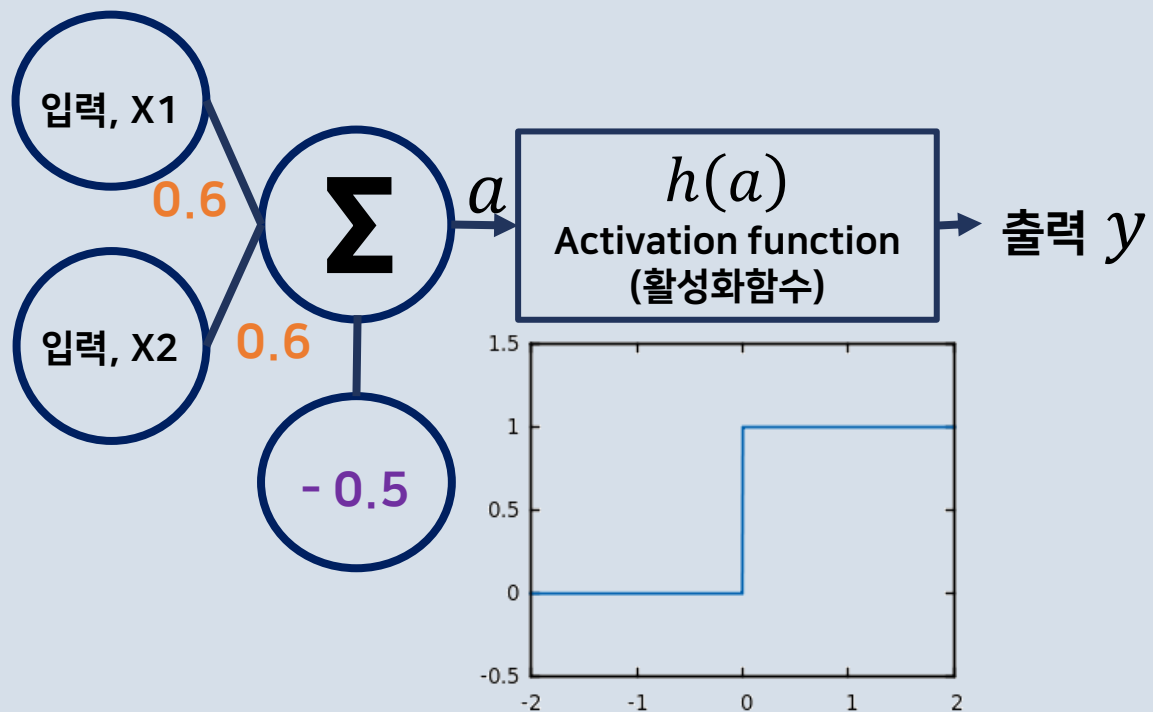
$$0.0 * 0.6 = 0.00$$

$$0.0 * 0.6 = 0.00$$

$$-0.50$$

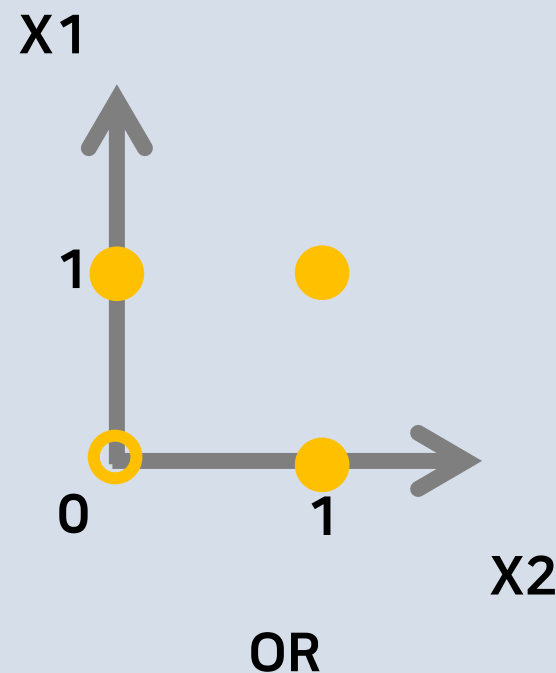
$$0.00 - 0.50 = -0.50$$

$$-0.50 \leq 0 \rightarrow 0$$

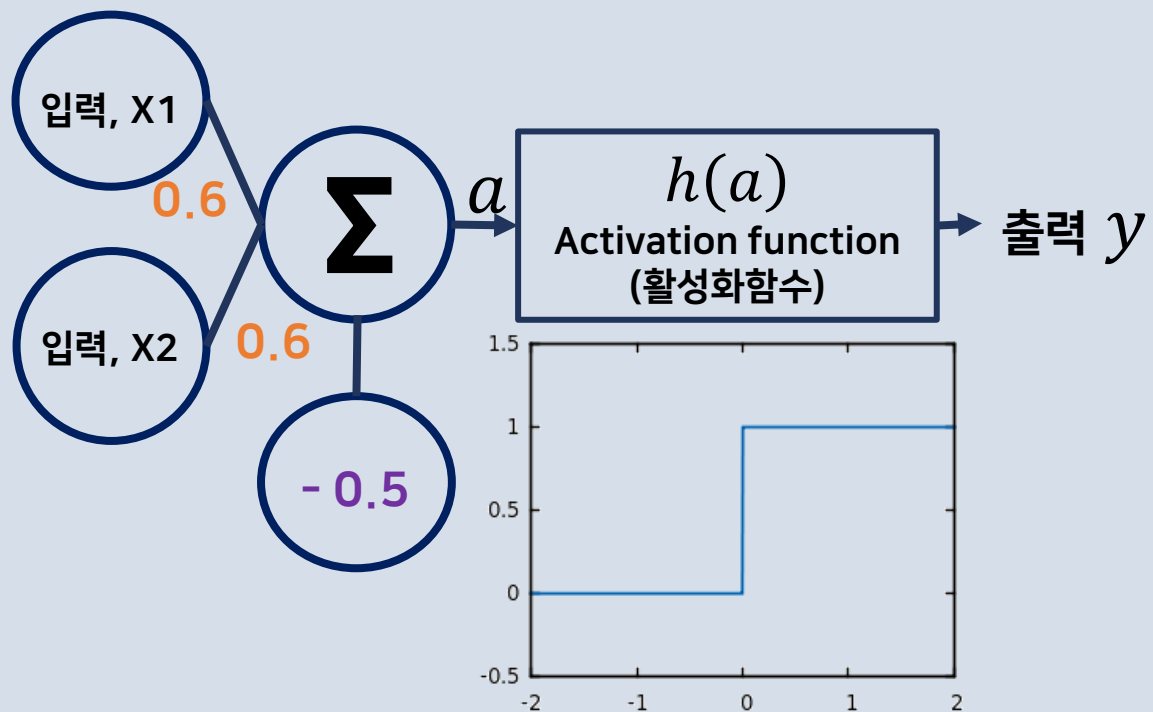


OR 게이트

X1	X2	OR
0	0	0
0	1	1
1	0	1
1	1	1

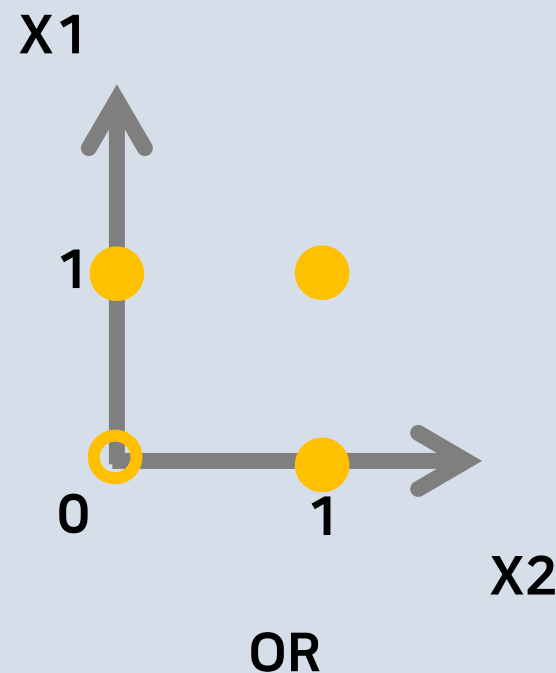


$$\begin{aligned}
 &0.0 * 0.6 = 0.00 \\
 &1.0 * 0.6 = 0.60 \\
 &\quad - 0.50 \\
 &\quad \rightarrow 0.60 - 0.50 = 0.10 \rightarrow 0.10 > 0 \rightarrow 1
 \end{aligned}$$



OR 게이트

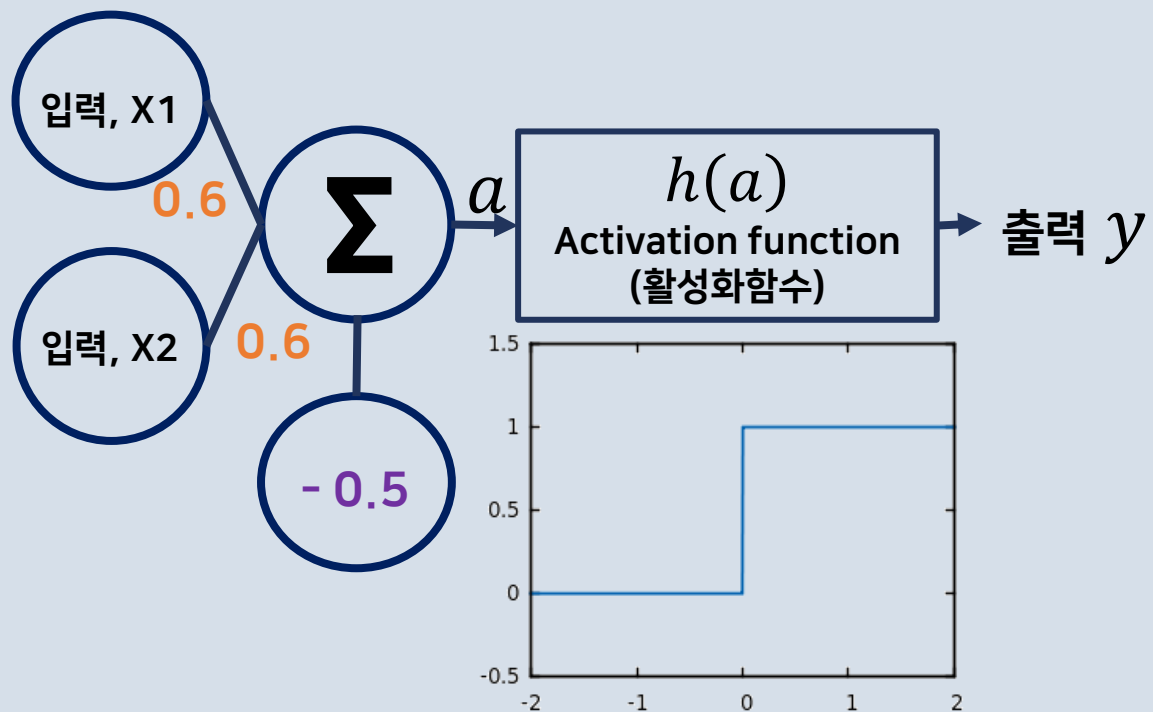
X1	X2	OR
0	0	0
0	1	1
1	0	1
1	1	1



$$1.0 * 0.6 = 0.60$$

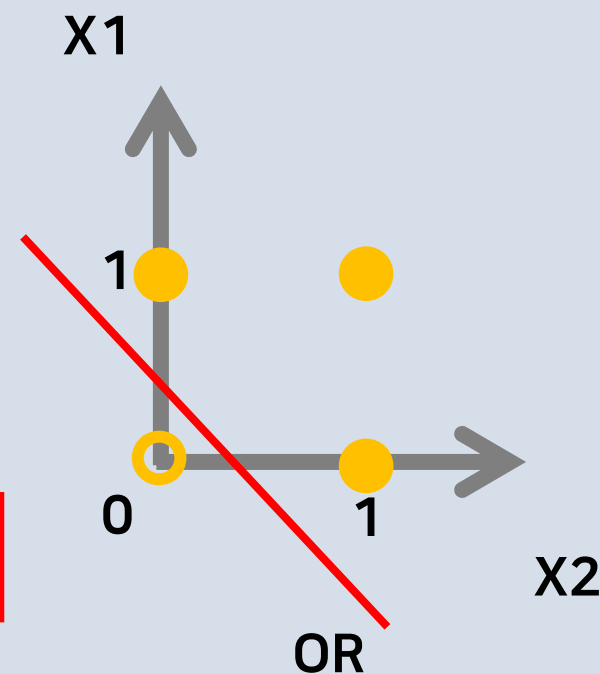
$$0.0 * 0.6 = 0.00$$

$$\begin{aligned} &\xrightarrow{\quad} 0.60 - 0.50 = 0.10 \quad \xrightarrow{\quad} 0.10 > 0 \quad \xrightarrow{\quad} 1 \\ &\quad - 0.50 \end{aligned}$$



OR 게이트

X1	X2	OR
0	0	0
0	1	1
1	0	1
1	1	1

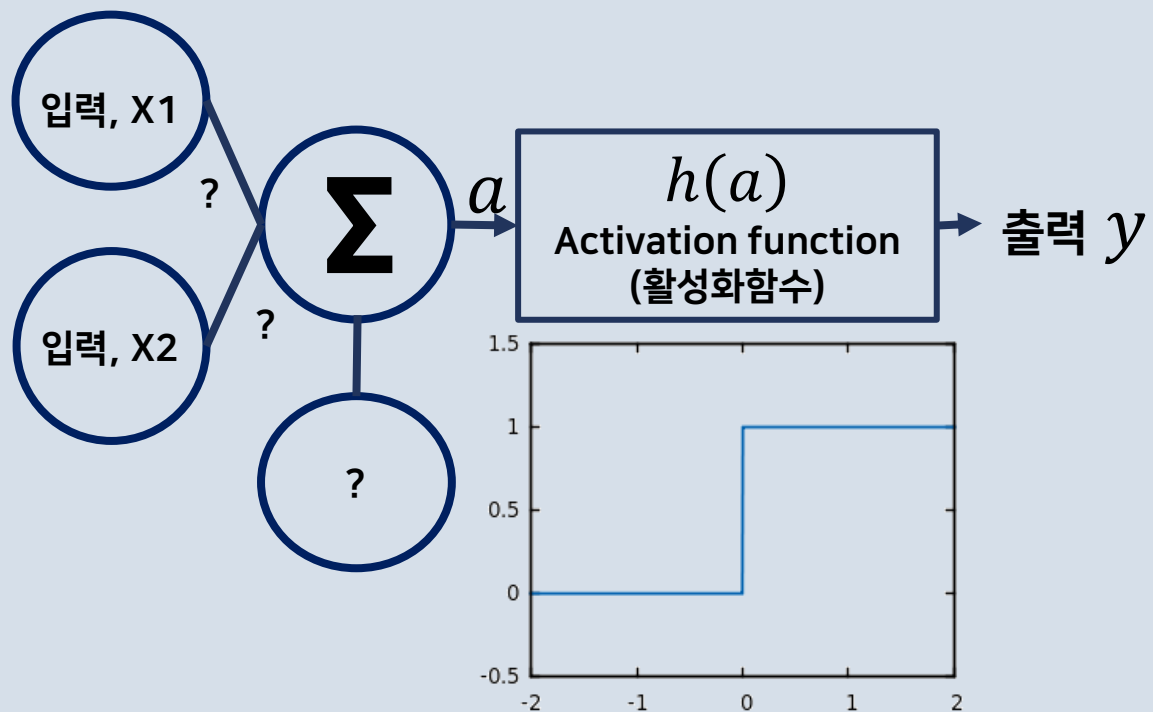


$$1.0 * 0.6 = 0.60$$

$$1.0 * 0.6 = 0.60$$

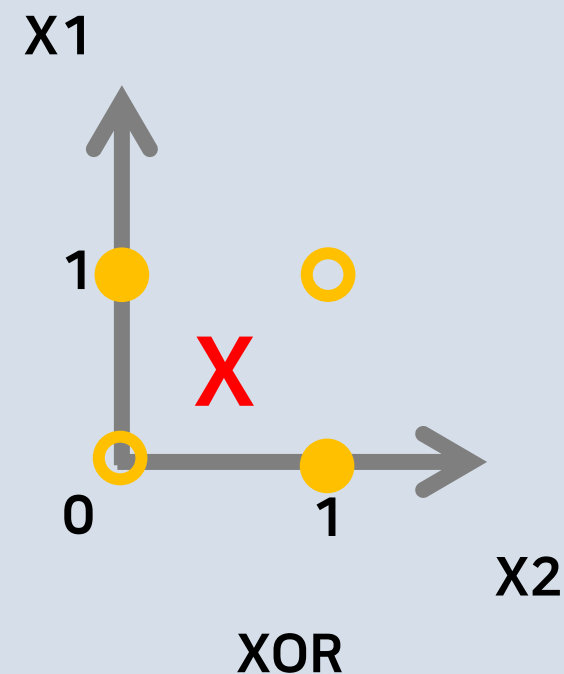
$$- 0.50$$

$$\rightarrow 1.20 - 0.50 = 0.70 \rightarrow 0.70 > 0 \rightarrow 1$$

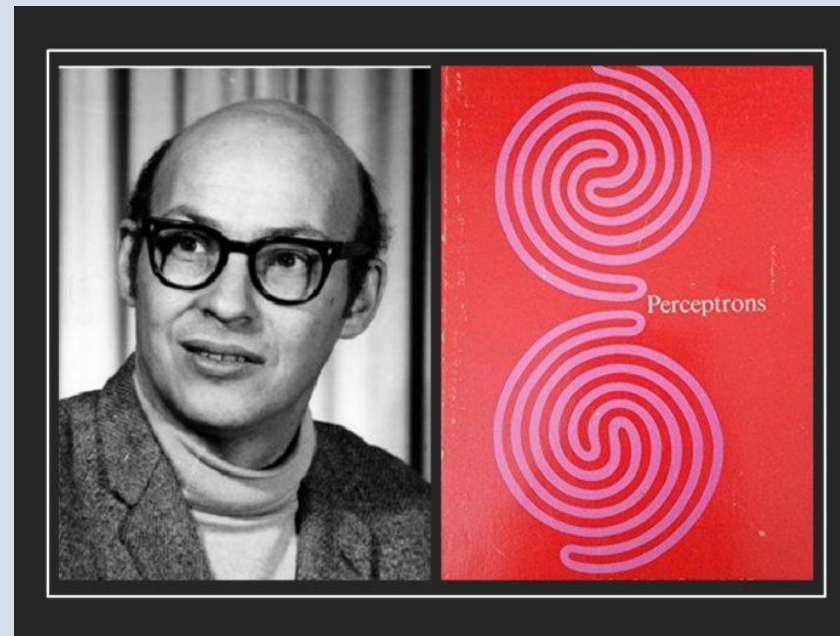


XOR 게이트

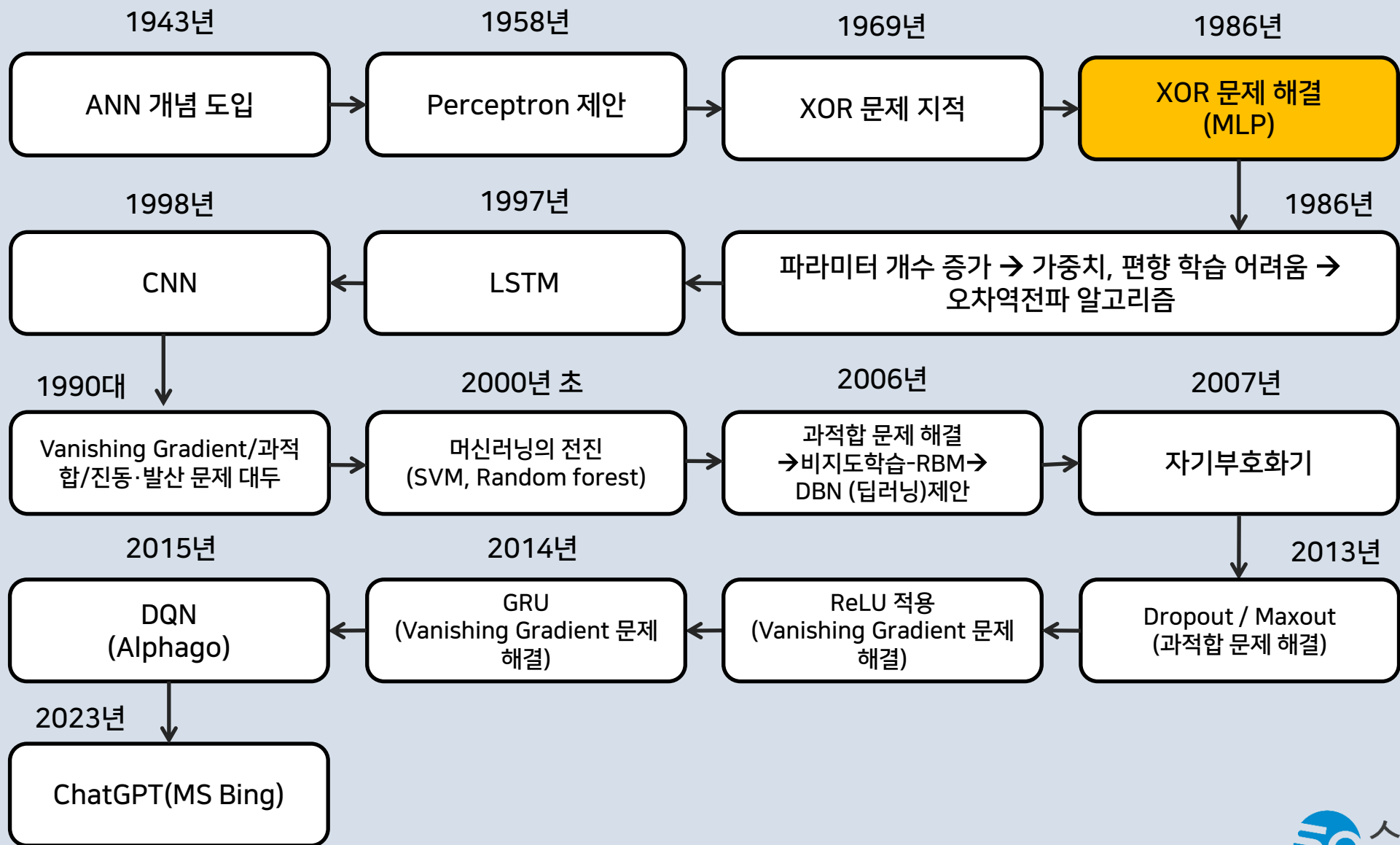
X1	X2	OR
0	0	0
0	1	1
1	0	1
1	1	0

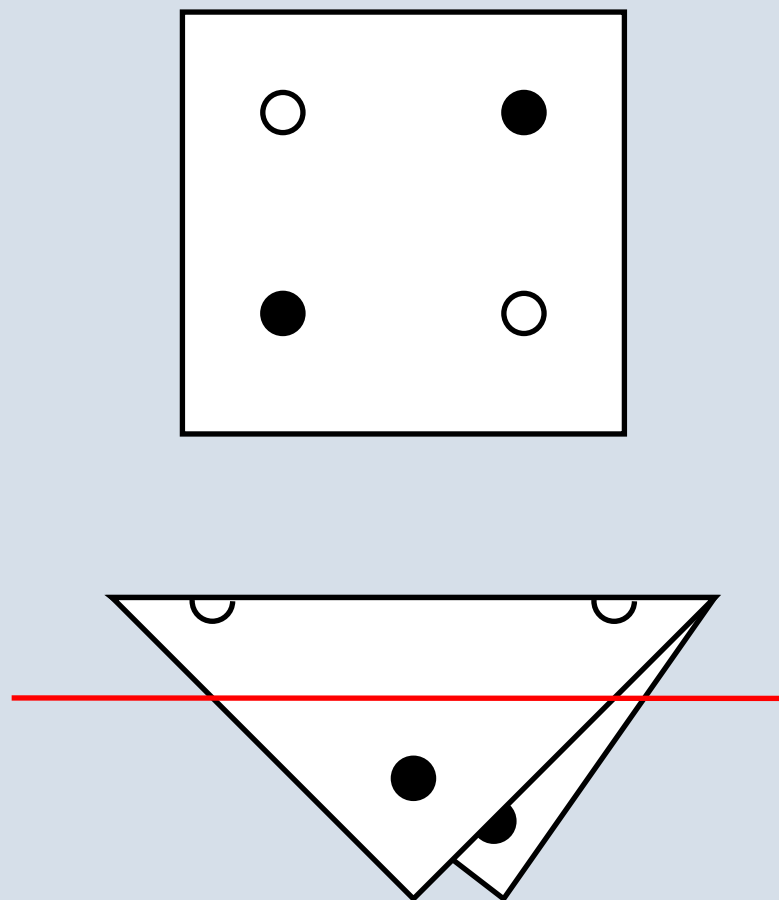
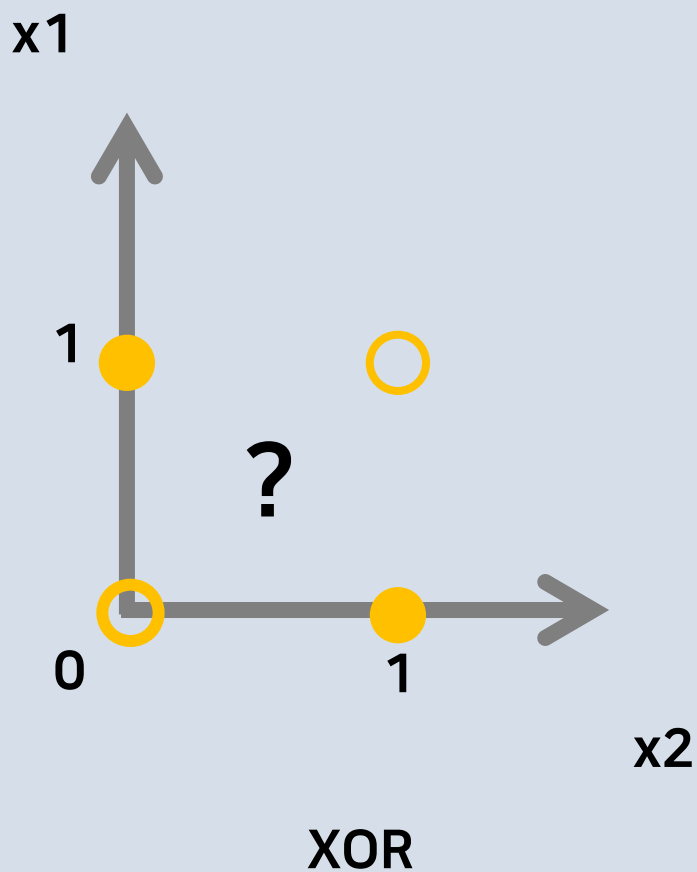


AND, OR는 해결이 가능하지만
간단한 XOR 문제를 해결 할 수 없었다.



출처 : 마빈 민스키와 퍼셉트론의 문제를 지적한 저서 <Perceptrons>

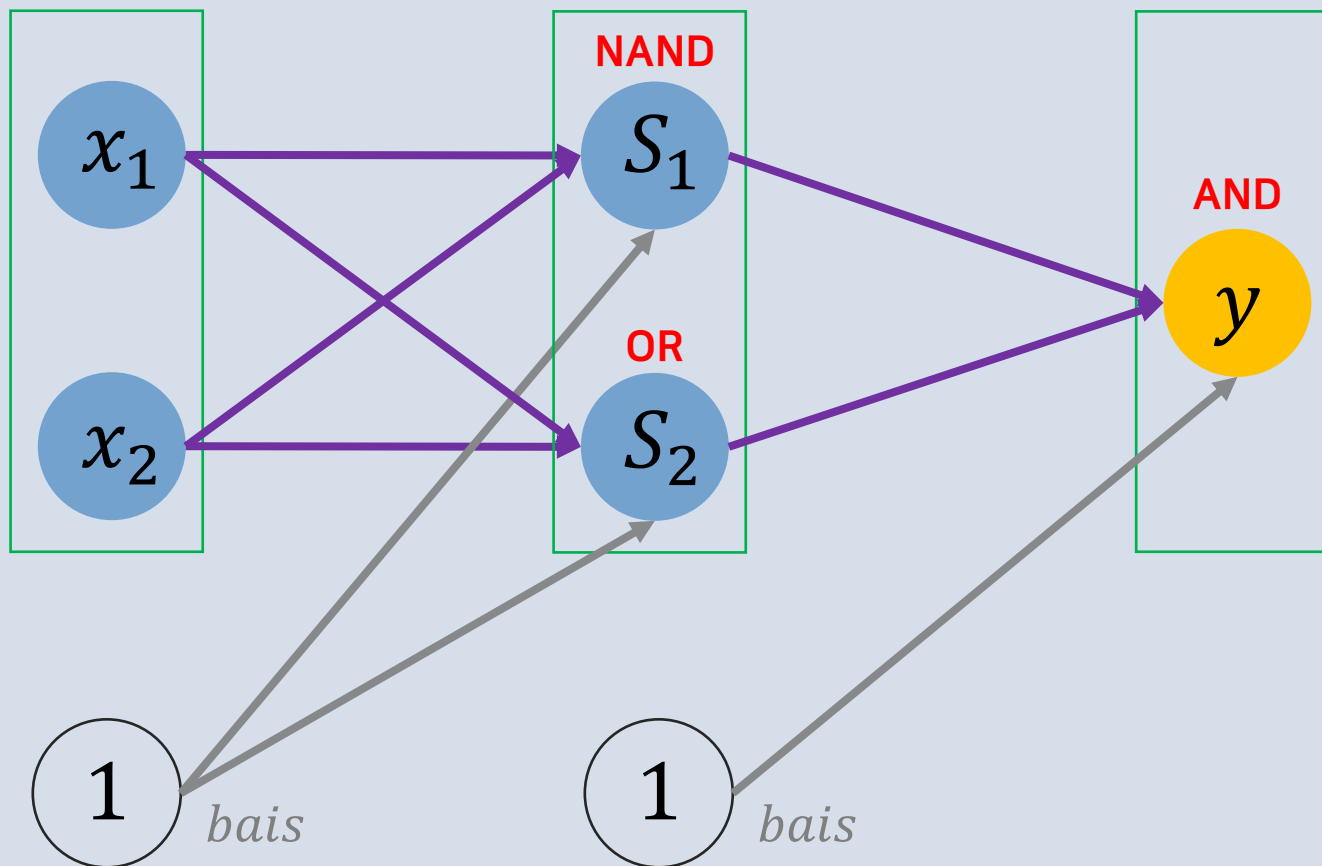




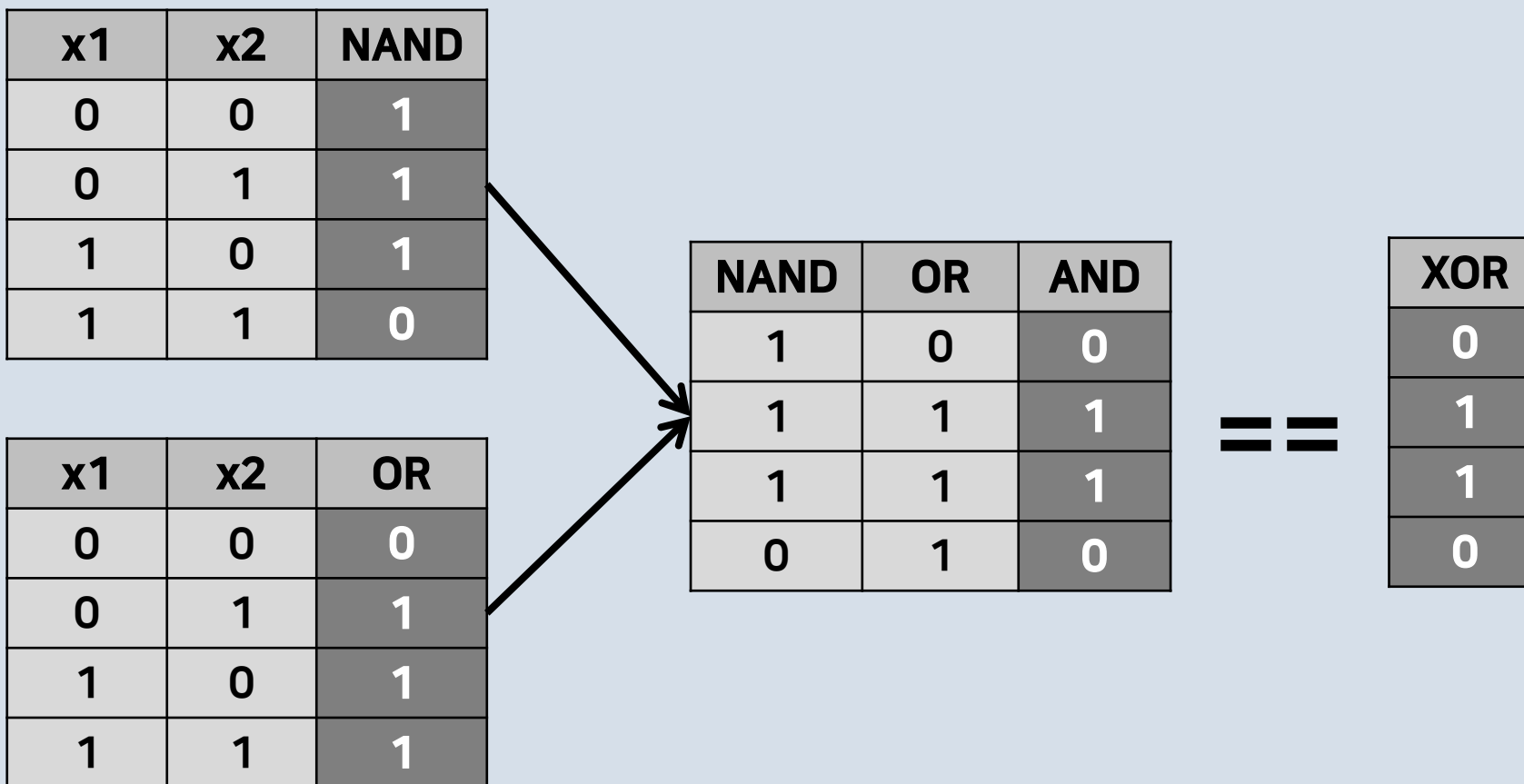
다층 퍼셉트론(Multi Layer Perceptron)

단층 퍼셉트론의 차원 수를 확장시켜
여러 개의 층으로 구성하여 만든 신경망.

입력층(input layer) 은닉층(hidden layer) 출력층(output layer)

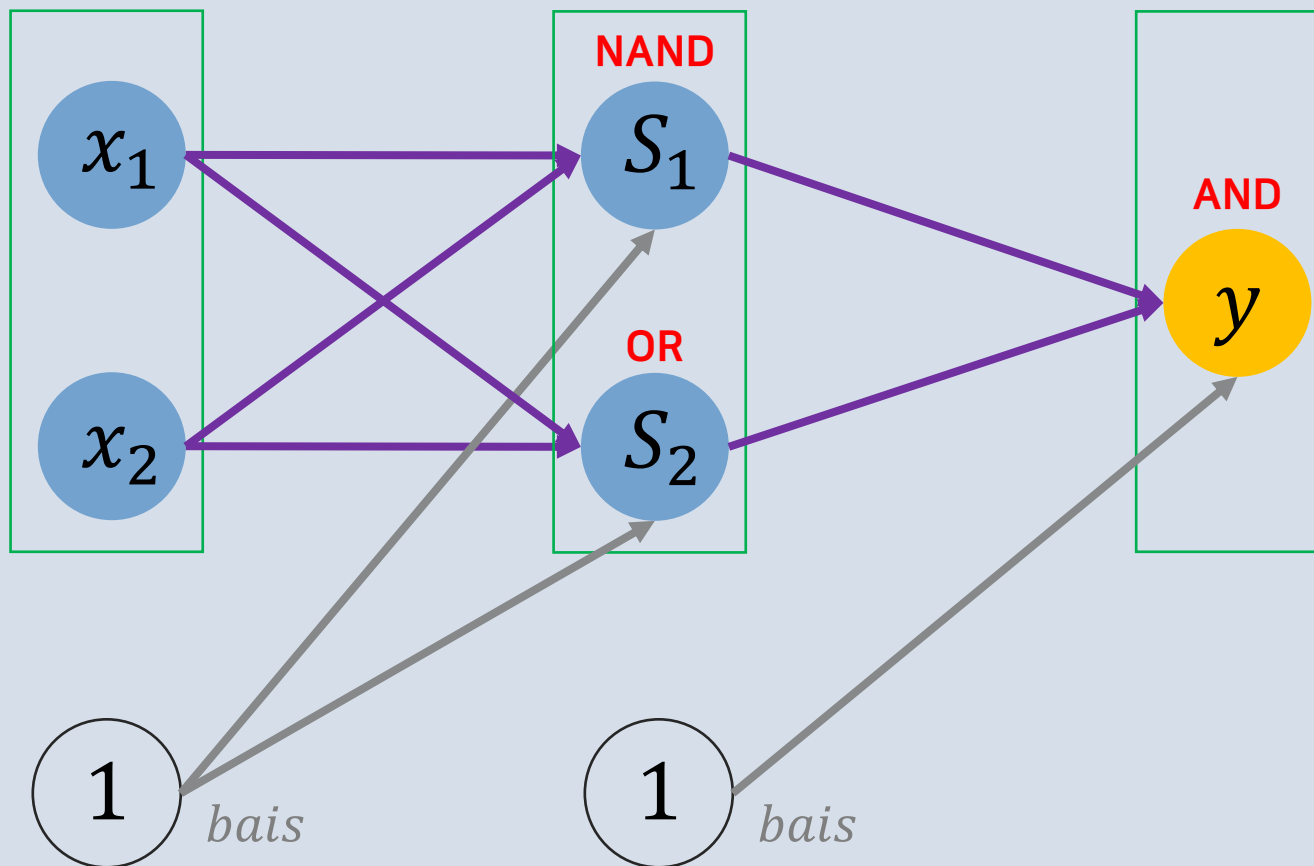


딥러닝 역사 - XOR 문제 해결(MLP)



다층 퍼셉트론(Multilayer Perceptron)

입력층(input layer) 은닉층(hidden layer) 출력층(output layer)

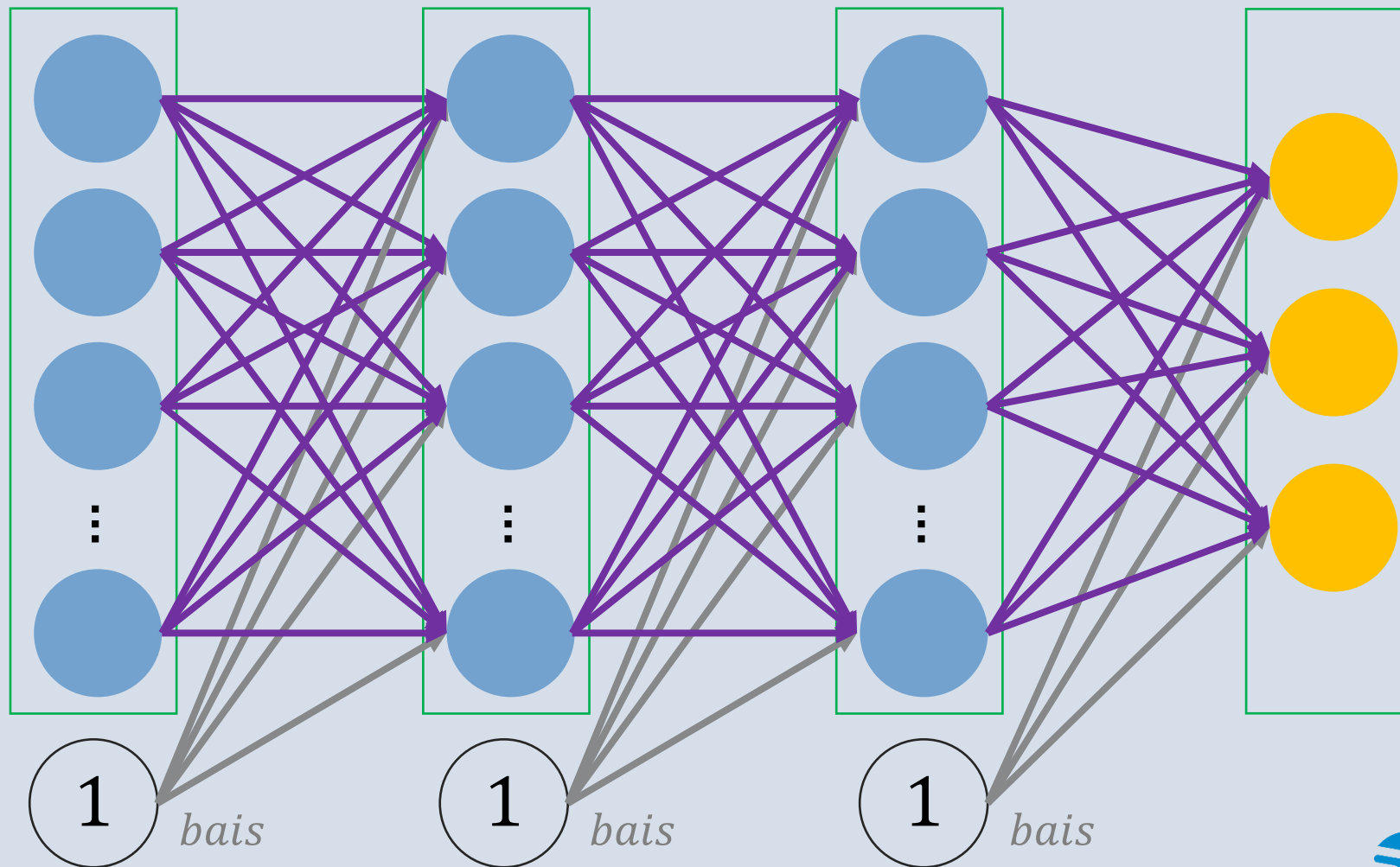


다층 퍼셉트론(Multilayer Perceptron)

입력층(input layer)

은닉층(hidden layer)

출력층(output layer)



- 한 번의 연산으로 해결되지 않는 문제를 해결할 수 있다.
- 단층에 비해 학습시간이 오래 걸린다.
- 모델(신경망)이 복잡해지고 가중치 파라미터가 많아 학습 시 과대적합되기 쉽다.

keras 맛보기 : 유방암 데이터 예측 실습